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A close-up photograph of several purple pea flowers (Pisum sativum) with green stems and leaves. The flowers are in various stages of bloom, showing the characteristic papilionaceous structure. The background is a soft, out-of-focus green.

14

Mendel and the Gene Idea

KEY CONCEPTS

- 14.1 Mendel used the scientific approach to identify two laws of inheritance
- 14.2 Probability laws govern Mendelian inheritance
- 14.3 Inheritance patterns are often more complex than predicted by simple Mendelian genetics
- 14.4 Many human traits follow Mendelian patterns of inheritance

Drawing from the Deck of Genes

- Every day we observe heritable variations (such as brown, green, or blue eyes) among individuals in a population.
 - These traits are transmitted from parents to offspring.
- **One possible explanation for heredity is a “blending” hypothesis.**
 - This hypothesis proposes that genetic material contributed by each parent mixes in a manner analogous to the way blue and yellow paints blend to make green.
 - With blending inheritance, a freely mating population would eventually give rise to a uniform population of individuals.
 - Everyday observations and the results of breeding experiments tell us that heritable traits do not blend to become uniform.
- **An alternative hypothesis, “particulate” inheritance, proposes that parents pass on discrete heritable units, genes, which retain their separate identities in offspring.**
 - Genes can be sorted and passed on, generation after generation, in undiluted form.
- Modern genetics began in an abbey garden, where a monk named **Gregor Mendel** documented a **particulate mechanism of inheritance**.



Mendel used the scientific approach to identify two laws of inheritance

Mendel took an experimental and quantitative approach.

- Mendel discovered the basic principles of heredity by breeding garden peas in carefully planned experiments, carried out several decades *before chromosomes were observed under the microscope*.
- Mendel grew up on a small farm in what is today the Czech Republic.
- In 1843, Mendel entered an Augustinian monastery.
- Mendel studied at the University of Vienna from 1851 to 1853, where he was influenced by a physicist who encouraged **experimentation and the application of mathematics to science** and by a botanist who stimulated Mendel's interest in **the causes of variation in plants**.
 - These influences came together in Mendel's experiments.
- After university, Mendel taught school and lived in the local monastery, where the monks had a long tradition of interest in the breeding of plants, including peas.
- Around 1857, Mendel began breeding garden peas to study inheritance.
- Pea plants have several advantages for genetic study.
 - Pea plants are available in many varieties that have distinct heritable features, or **characters**, with different variant **traits**.
 - Peas have a short generation time, and each mating produces many offspring.

Mendel used the scientific approach to identify two laws of inheritance

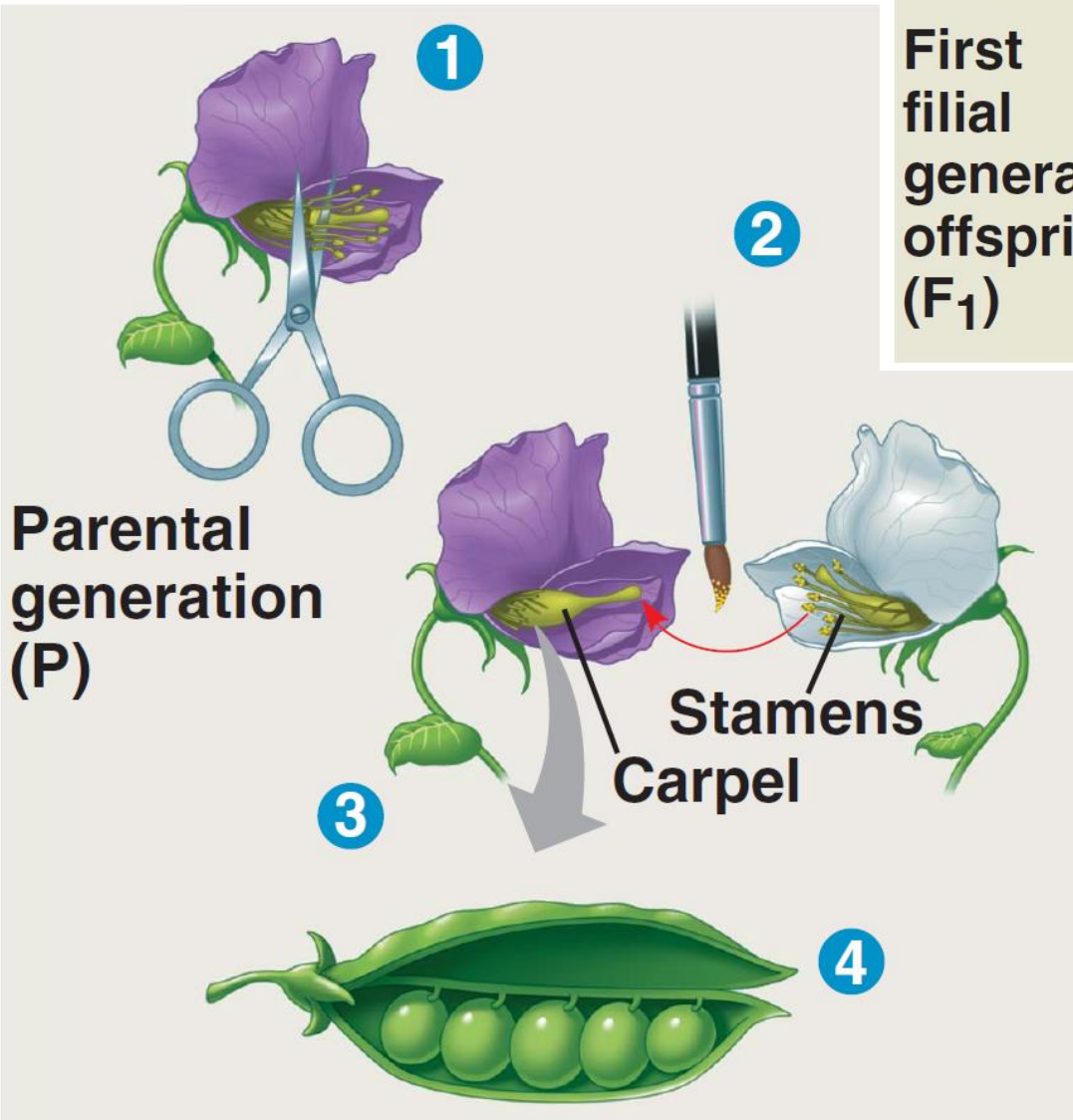
Mendel took an experimental and quantitative approach.

- Mendel was able to strictly control the mating between his pea plants.
 - Each pea plant has **male (stamens)** and **female (carpel)** sexual organs.
 - In nature, pea plants typically self-fertilize, fertilizing ova with the sperm nuclei from their own pollen.
 - Mendel could also use pollen from another plant for cross-pollination.
- Mendel tracked only those characters that varied in an “either-or” manner, rather than a “more-or-less” manner.
 - For example, he worked with flowers that were either purple or white.
 - He avoided traits such as seed weight, which varied on a continuum.
- Meiosis in plants produces spores, not gametes.
 - In flowering plants like the pea, each spore develops into a microscopic haploid gametophyte that contains only a few cells and is located on the parent plant.
 - The gametophyte produces sperm, in pollen grains, and eggs, in the carpel.
 - *For simplicity, the gametophyte stage will not be included in this discussion of plant breeding.*

Mendel used the scientific approach to identify two laws of inheritance

Mendel took an experimental and quantitative approach.

Technique



Results

First filial generation offspring (F₁)



Mendel used the scientific approach to identify two laws of inheritance

Mendel took an experimental and quantitative approach.

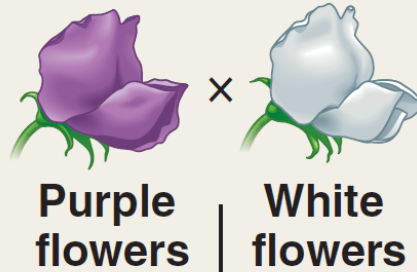
- Mendel started his experiments with varieties that were **true-breeding**.
 - When true-breeding plants self-pollinate, all their offspring have the same traits as their parents.
- In a typical breeding experiment, Mendel would cross-pollinate (**hybridize**) two contrasting, true-breeding pea varieties.
 - The true-breeding parents are the **P (parental) generation**, and their hybrid offspring are the **F₁ (first filial) generation**.
- Mendel would then allow the F₁ hybrids to self-pollinate to produce an **F₂ (second filial) generation**.
- It was mainly Mendel's quantitative analysis of F₂ plants that led him to deduce two fundamental principles of heredity: the law of segregation and the law of independent assortment.

Mendel used the scientific approach to identify two laws of inheritance

Mendel took an experimental and quantitative approach.

Experiment

P Generation
(true-breeding
parents)



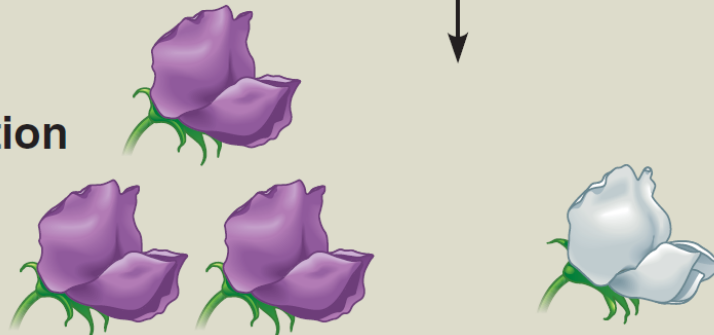
F₁ Generation
(hybrids)



All plants had purple flowers

Self- or cross-pollination

F₂ Generation



705 purple-flowered
plants

224 white-flowered
plants

Mendel used the scientific approach to identify two laws of inheritance

The Law of Segregation

- If the blending hypothesis were correct, the F_1 hybrids from a cross between purple-flowered and white-flowered pea plants would have pale purple flowers.
 - Instead, the F_1 hybrids all have purple flowers, just as purple as their purple-flowered parents.
- When Mendel allowed the F_1 plants to self-fertilize, the F_2 generation included both purple-flowered and white-flowered plants.
 - The white trait, absent in the F_1 generation, reappeared in the F_2 .
- Mendel used very large sample sizes and kept accurate records of his results.
 - Mendel recorded 705 purple-flowered F_2 plants and 224 white-flowered F_2 plants.
 - The cross produced a ratio of three purple flowers to one white flower in the F_2 offspring.
- Mendel reasoned that the heritable factor for white flowers was present in the F_1 plants but did not affect flower color.
 - Purple flower color is a *dominant* trait, and white flower color is a *recessive* trait.
 - The reappearance of white-flowered plants in the F_2 generation indicated that the heritable factor for the white trait was not diluted or lost by coexisting with the purple-flower factor in F_1 hybrids.

Mendel used the scientific approach to identify two laws of inheritance

Table 14.1 The Results of Mendel's F ₁ Crosses for Seven Characters in Pea Plants					
Character	Dominant Trait	×	Recessive Trait	F ₂ Generation Dominant: Recessive	Ratio
Flower color	Purple 	×	White 	705:224	3.15:1
Seed color	Yellow 	×	Green 	6,022:2,001	3.01:1
Seed shape	Round 	×	Wrinkled 	5,474:1,850	2.96:1
Pod shape	Inflated 	×	Constricted 	882:299	2.95:1
Pod color	Green 	×	Yellow 	428:152	2.82:1
Flower position	Axial 	×	Terminal 	651:207	3.14:1
Stem length	Tall 	×	Dwarf 	787:277	2.84:1

Mendel used the scientific approach to identify two laws of inheritance

The Law of Segregation

- Mendel found similar 3:1 ratios of two traits in F_2 offspring when he conducted crosses for six other characters, each represented by two different traits.
 - For example, when Mendel crossed two true-breeding varieties, one producing **round seeds** and the other producing **wrinkled seeds**, all the F_1 offspring had round seeds.
 - In the F_2 plants, about 75% of the seeds were round and 25% were wrinkled.
- Mendel developed a hypothesis to explain these results that consisted of four related ideas. We will explain each idea with the modern understanding of genes and chromosomes.

1. Alternative versions of genes account for variations in inherited characters.

- The gene for flower color in pea plants exists in two versions, one for purple flowers and one for white flowers.
- *These alternative versions of a gene are called **alleles**.*
- Each gene resides at a specific locus on a specific chromosome. The DNA at that locus can vary in its sequence of nucleotides.
- The purple-flower and white-flower alleles are two DNA sequence variations at the **flower-color locus**.

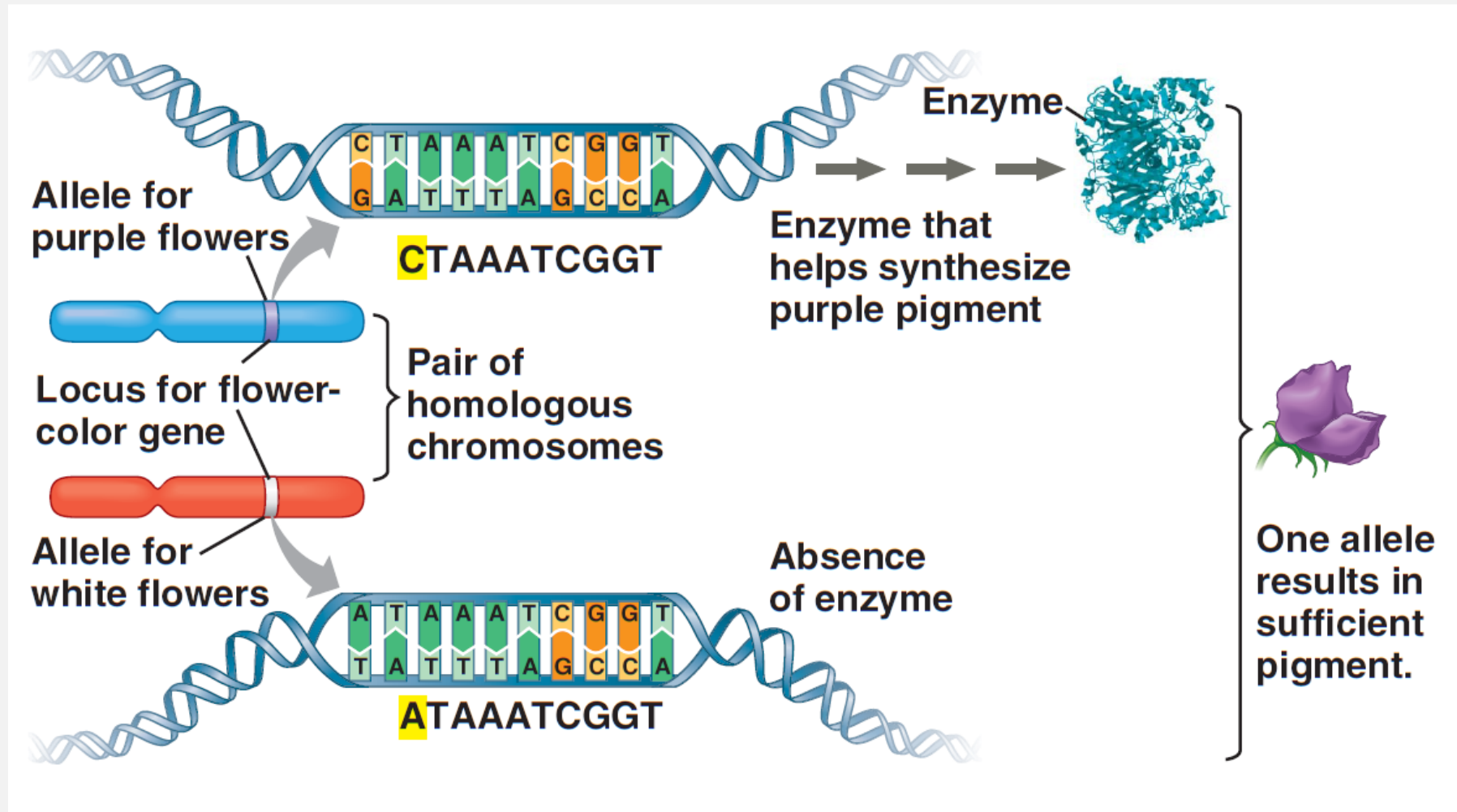
Mendel used the scientific approach to identify two laws of inheritance

The Law of Segregation

2. *For each character, an organism inherits two copies of a gene, one from each parent.*
 - A diploid organism inherits one set of chromosomes from each parent.
 - Each diploid organism has a pair of homologous chromosomes and, therefore, two copies of each gene. These are also called alleles of that gene.
 - These homologous loci may be identical, as in the true-breeding plants of the P generation.
 - Alternatively, the two alleles may differ, as in the F_1 hybrids.
3. *If the two alleles at a locus differ, then one, the dominant allele, determines the organism's appearance. The other, the recessive allele, has no noticeable effect on the organism's appearance.*
 - In the flower-color example, the F_1 plants inherited a purple-flower allele from one parent and a white-flower allele from the other.
 - The plants had purple flowers because the allele for that trait is dominant.

Mendel used the scientific approach to identify two laws of inheritance

The Law of Segregation



Mendel used the scientific approach to identify two laws of inheritance

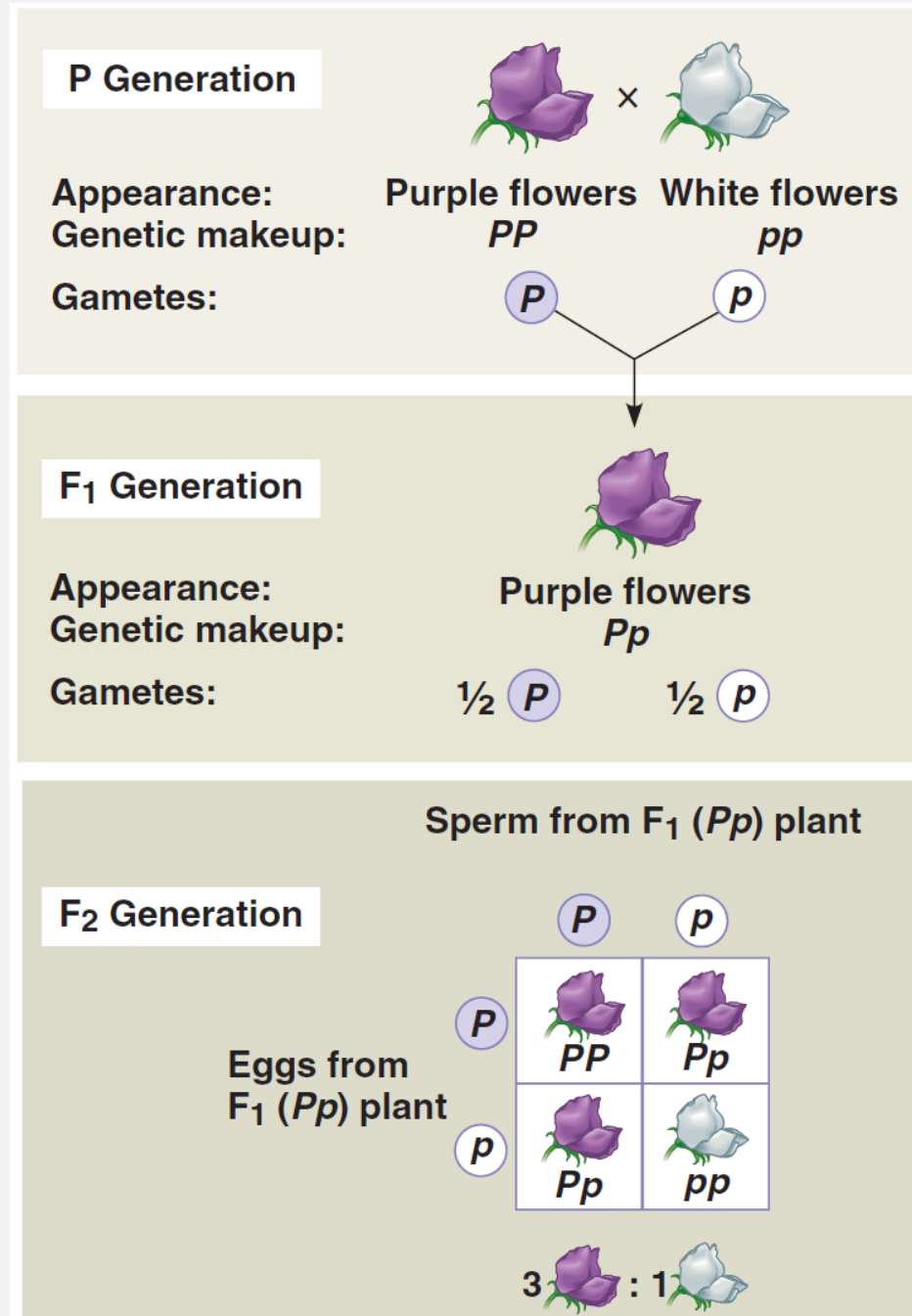
The Law of Segregation

4. *Mendel's law of segregation states that the two alleles for a heritable character segregate (separate) during gamete production and end up in different gametes.*

- This segregation of alleles corresponds to the distribution of homologous chromosomes to different gametes in meiosis.
- If an organism has two identical alleles for a particular character, then that allele is present as a single copy in all gametes.
- If different alleles are present, then 50% of the gametes will receive one allele and 50% will receive the other.
- Mendel's law of segregation accounts for the 3:1 ratio that he observed in the F_2 generation.
- The F_1 hybrids produce two classes of gametes, half with the purple-flower allele and half with the white-flower allele.
- During self-pollination, the gametes of these two classes unite randomly to produce four equally likely combinations of sperm and ovum.
- A **Punnett square** may be used to predict the results of a genetic cross between individuals of known genotype.
- For the flower-color example, we can use a capital letter to symbolize the dominant allele and a lowercase letter to symbolize the recessive allele.
 - P is the purple-flower allele, and p is the white-flower allele.

Mendel used the scientific approach to identify two laws of inheritance

The Law of Segregation



Mendel used the scientific approach to identify two laws of inheritance

The Law of Segregation

- What will be the physical appearance of the F_2 offspring?
 - One in four F_2 offspring will inherit two white-flower alleles and produce white flowers.
 - Half of the F_2 offspring will inherit one white-flower allele and one purple-flower allele and produce purple flowers.
 - One in four F_2 offspring will inherit two purple-flower alleles and produce purple flowers.
- Mendel's model accounts for the 3:1 ratio in the F_2 generation.

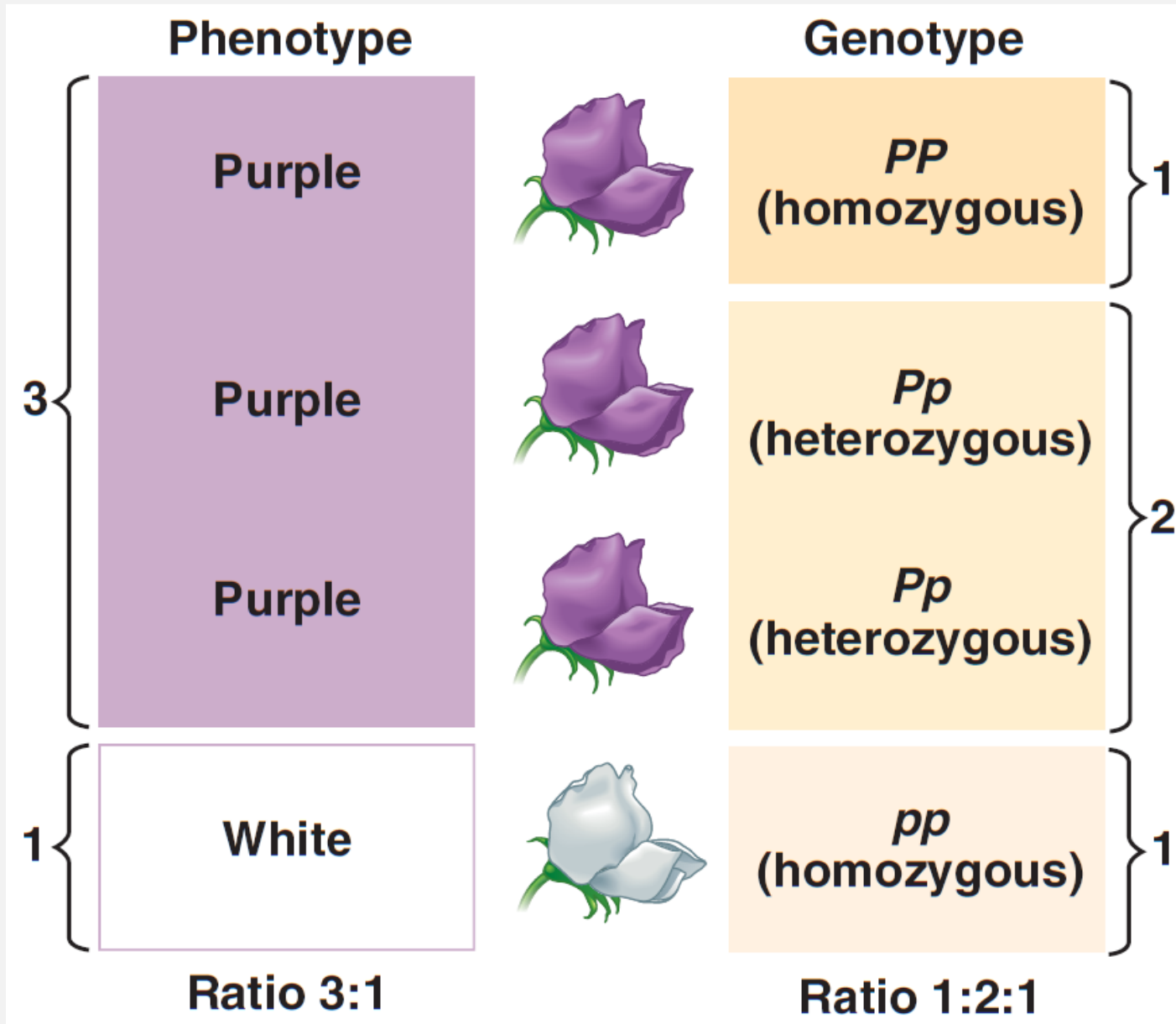
Mendel used the scientific approach to identify two laws of inheritance

Useful Genetic Vocabulary

- An organism with two identical alleles for a character is **homozygous** for the gene controlling that character.
- An organism with two different alleles for a gene is **heterozygous** for that gene.
- An organism's observable traits are called its **phenotype**.
 - "Phenotype" refers to physiological traits as well as traits directly related to appearance.
- An organism's genetic makeup is called its **genotype**.
- Two organisms can have the same phenotype but different genotypes if one is homozygous dominant and the other is heterozygous.
 - PP and Pp plants have the same phenotype (purple flowers) but different genotypes (homozygous dominant and heterozygous).
- For flower color in peas, the only individuals with white flowers are those that are homozygous recessive (pp) for the flower-color gene.

Mendel used the scientific approach to identify two laws of inheritance

Useful Genetic Vocabulary



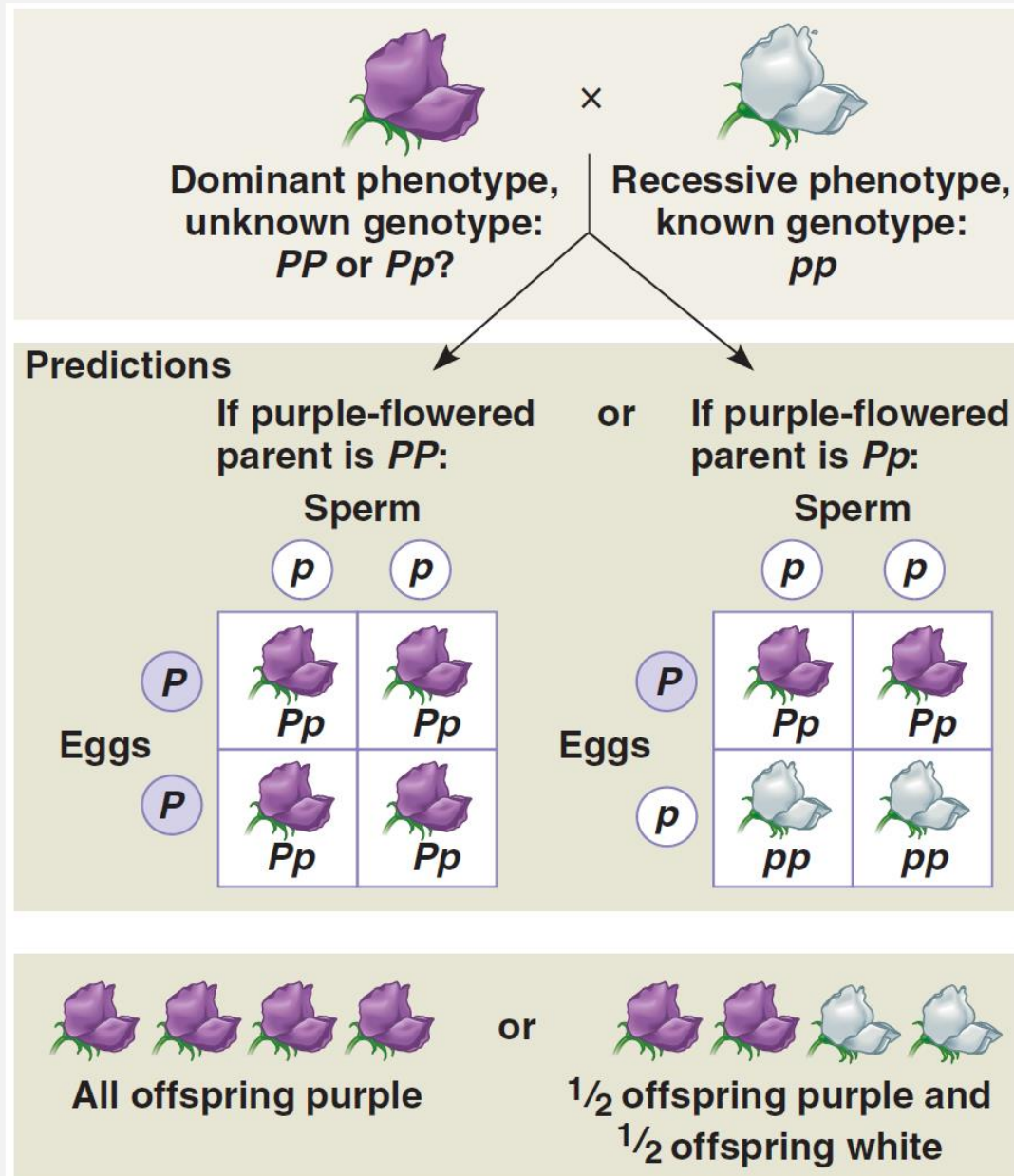
Mendel used the scientific approach to identify two laws of inheritance

A testcross can be used to determine the genotype of an individual with the dominant phenotype.

- How can we determine the genotype of an individual that has the dominant phenotype, e.g., a pea plant with purple flowers?
 - The organism must have one dominant allele but could be homozygous dominant or heterozygous.
- The answer is to carry out a **testcross**.
 - The mystery individual is bred with a homozygous recessive individual.
 - If any of the offspring display the recessive phenotype, the mystery parent must be heterozygous.

Mendel used the scientific approach to identify two laws of inheritance

A testcross can be used to determine the genotype of an individual with the dominant phenotype.



Mendel used the scientific approach to identify two laws of inheritance

The Law of Independent Assortment

- Mendel's first experiments followed only a *single* character, such as flower color.
 - All the F_1 progeny produced in these crosses were **monohybrids**, heterozygous for one character.
 - A cross between two heterozygotes is a *monohybrid cross*.
- Mendel identified the second law of inheritance by following *two* characters at the same time.
- In one such **dihybrid cross**, Mendel studied the inheritance of seed color and seed shape.
 - The allele for yellow seeds (Y) is dominant to the allele for green seeds (y).
 - The allele for round seeds (R) is dominant to the allele for wrinkled seeds (r).
- Mendel crossed true-breeding plants that had yellow, round seeds ($YYRR$) with true-breeding plants that had green, wrinkled seeds ($yyrr$).
- The F_1 plants are **dihybrid** individuals that are heterozygous for two characters ($YyRr$).
- One possible hypothesis is that the two characters are transmitted from parents to offspring as a package.
 - In this case, the Y and R alleles and the y and r alleles would stay together.
 - If this were the case, the F_1 offspring would produce yellow, round seeds.
 - The F_2 offspring would produce two phenotypes (yellow + round; green + wrinkled) in a 3:1 ratio, just like a monohybrid cross.
 - This was not consistent with Mendel's results.

Mendel used the scientific approach to identify two laws of inheritance

The Law of Independent Assortment

- An alternative hypothesis is that the two pairs of alleles segregate independently of each other.
 - The presence of a specific allele for one trait in a gamete has no impact on the presence of a specific allele for the second trait.
 - In our example, the F_1 offspring would still produce yellow, round seeds.
 - When the F_1 offspring produced gametes, genes would be packaged into gametes with all possible allelic combinations.
 - Four classes of gametes (YR , Yr , yR , and yr) would be produced in equal amounts.
 - When sperm with four classes of alleles and ova with four classes of alleles combine, there are 16 equally probable ways in which the alleles can combine in the F_2 generation.
 - These combinations produce four distinct phenotypes in a 9:3:3:1 ratio.
 - This was consistent with Mendel's experimental results.
- Mendel repeated the dihybrid cross experiment for other pairs of characters and always observed a 9:3:3:1 phenotypic ratio in the F_2 generation.
 - Each character appeared to be inherited independently.
 - If you follow just one character in these crosses, you will observe a 3:1 F_2 ratio, just as if this were a monohybrid cross.

Mendel used the scientific approach to identify two laws of inheritance

The Law of Independent Assortment

Experiment

P Generation

YYRR  **yyrr** 

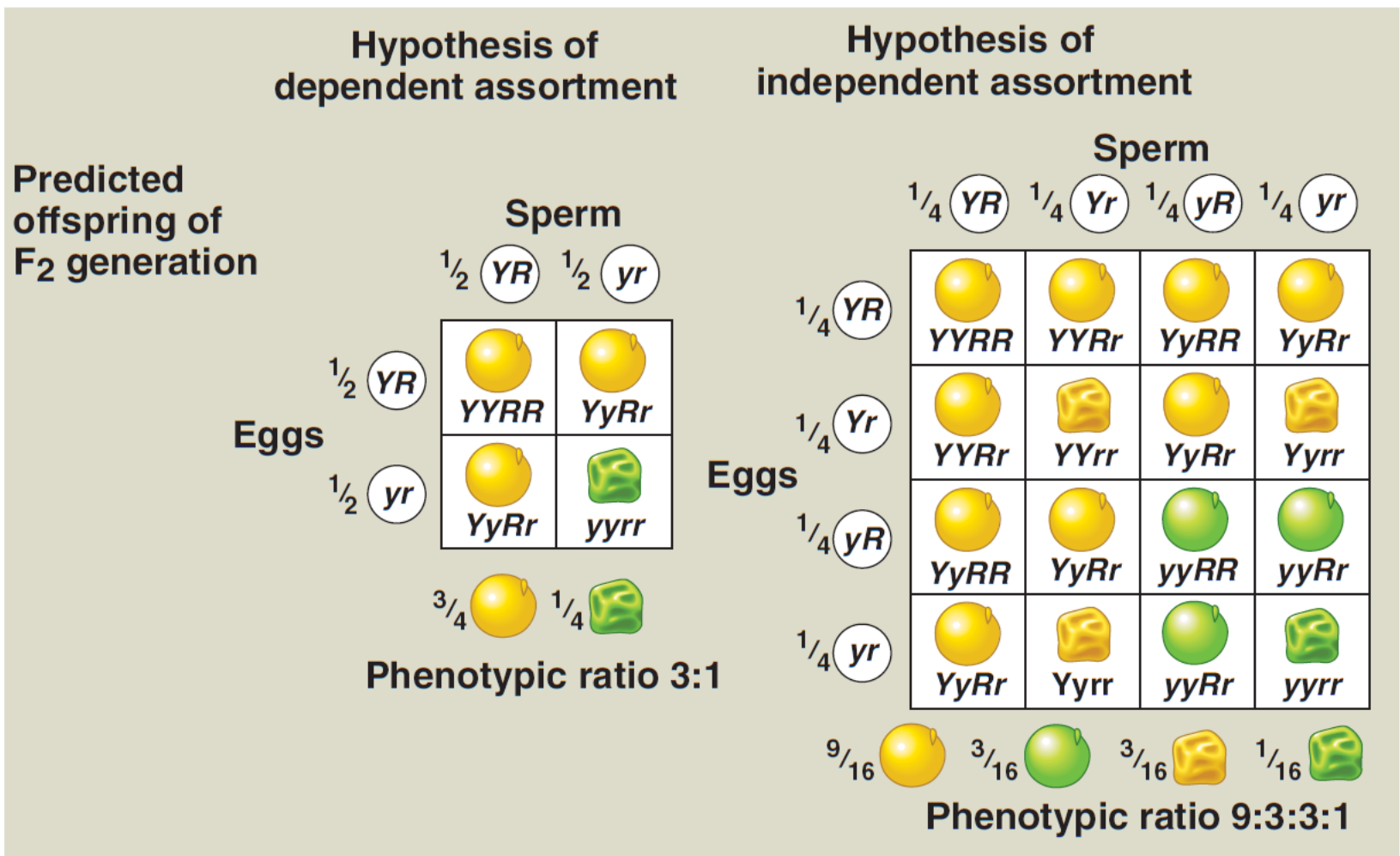
Gametes **YR** × **yr**

F₁ Generation

 **YyRr**

Mendel used the scientific approach to identify two laws of inheritance

The Law of Independent Assortment

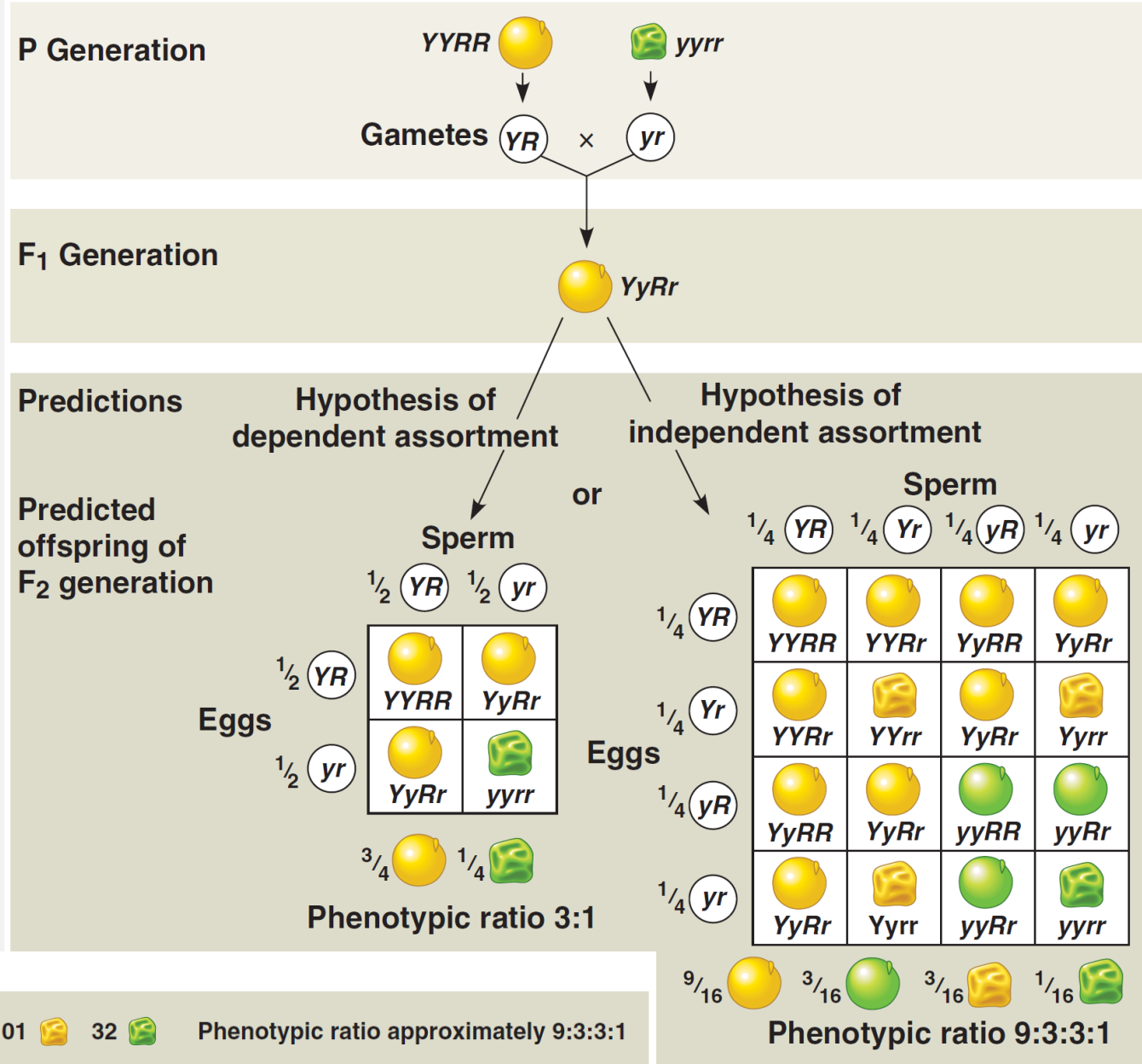


Results

315  108  101  32  Phenotypic ratio approximately 9:3:3:1

Mendel used the scientific approach to identify two laws of inheritance

The Law of Independent Assortment



Mendel used the scientific approach to identify two laws of inheritance

The Law of Independent Assortment

- The independent assortment of each pair of alleles during gamete formation is called Mendel's **law of independent assortment**: Each pair of alleles segregates independently during gamete formation.
- Strictly speaking, this law applies only to genes located on different, nonhomologous chromosomes.
 - Genes located near each other on the same chromosome tend to be inherited together and have more complex inheritance patterns than those predicted for the law of independent assortment.

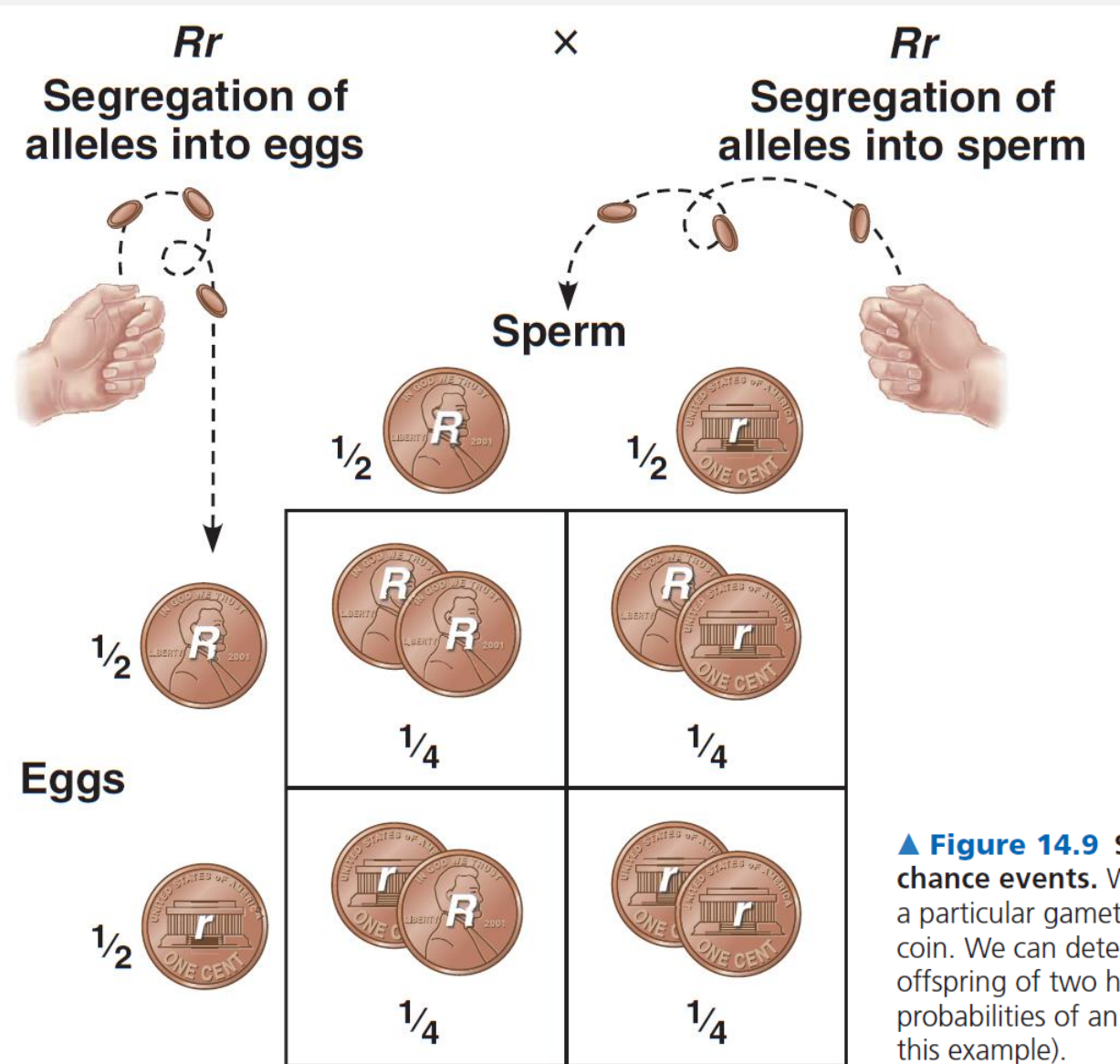
Probability laws govern Mendelian inheritance

- Mendel's laws of segregation and independent assortment reflect the same laws of probability that apply to tossing coins or rolling dice.
- Values of probability range from 0 (an event with no chance of occurring) to 1 (an event that is certain to occur).
 - The probability of tossing heads with a normal coin is $1/2$.
 - The probability of rolling a 3 with a six-sided die is $1/6$, and the probability of rolling any other number is $1 - 1/6 = 5/6$.
- The outcome of one coin toss has no impact on the outcome of the next toss. Each toss is an independent event, just like the distribution of alleles into gametes.
 - Like a coin toss, each ovum from a heterozygous parent has a $1/2$ chance of carrying the dominant allele and a $1/2$ chance of carrying the recessive allele.
 - The same probabilities apply to the sperm.

Probability laws govern Mendelian inheritance

- We can use the **multiplication rule** to determine the probability that two or more independent events will occur together in some specific combination.
 - Compute the probability of each independent event and then multiply the individual probabilities to obtain the overall probability of these events occurring together.
 - The probability that two coins tossed at the same time will both land heads up is $1/2 \times 1/2 = 1/4$.
 - Similarly, the probability that a heterozygous pea plant (Pp) will self-fertilize to produce a white-flowered offspring (pp) is the probability that a sperm with a white allele will fertilize an ovum with a white allele. This probability is $1/2 \times 1/2 = 1/4$.
- We can use the **addition rule** to determine the probability that an F_2 plant from a monohybrid cross will be heterozygous rather than homozygous.
 - The probability of an event that can occur in two or more mutually exclusive ways is the sum of the individual probabilities of those ways.
 - The probability of obtaining an F_2 heterozygote by combining the dominant allele from the egg and the recessive allele from the sperm is $1/4$.
 - The probability of combining the recessive allele from the egg and the dominant allele from the sperm also $1/4$.
 - Using the rule of addition, we can calculate the probability of an F_2 heterozygote as $1/4 + 1/4 = 1/2$.

Probability laws govern Mendelian inheritance



Probability governs Mendelian Inheritance

▲ **Figure 14.9** Segregation of alleles and fertilization as chance events. When a heterozygote (Rr) forms gametes, whether a particular gamete ends up with an R or an r is like the toss of a coin. We can determine the probability for any genotype among the offspring of two heterozygotes by multiplying together the individual probabilities of an egg and sperm having a particular allele (R or r in this example).

Probability laws govern Mendelian inheritance

- The rule of multiplication applies to dihybrid crosses.
 - For a heterozygous parent ($YyRr$), the probability of producing a YR gamete is $1/2 \times 1/2 = 1/4$.
 - We can now predict the probability of a particular F_2 genotype ***without constructing a 16-part Punnett square***.
 - The probability that an F_2 plant from heterozygous parents will have a $YYRR$ genotype is $1/16$ ($1/4$ chance for a YR ovum $\times$ $1/4$ chance for a YR sperm).
- We can combine the rules of multiplication and addition to solve complex problems in Mendelian genetics.
- Let's determine the probability of an offspring having two recessive phenotypes for *at least two* of three traits resulting from a trihybrid cross between pea plants that are $PpYyRr$ and $Ppyyrr$.
 - Five possible genotypes result in this condition: $ppyyRr$, $ppYyrr$, $Ppyyrr$, $PPyyrr$, and $ppyyrr$.
 - We can use the rule of multiplication to calculate the probability for each of these genotypes and then use the rule of addition to pool the probabilities for finding at least two recessive traits.

Probability laws govern Mendelian inheritance

- The probability of producing a $ppyyRr$ offspring:
 - The probability of producing $pp = 1/2 \times 1/2 = 1/4$.
 - The probability of producing $yy = 1/2 \times 1 = 1/2$.
 - The probability of producing $Rr = 1/2 \times 1 = 1/2$.
 - Therefore, the probability of all three being present ($ppyyRr$) in one offspring is $1/4 \times 1/2 \times 1/2 = 1/16$.
 - For $ppYyrr$: $1/4 \times 1/2 \times 1/2 = 1/16$.
 - For $Ppyyrr$: $1/2 \times 1/2 \times 1/2 = 1/8$ or $2/16$.
 - For $PPyyrr$: $1/4 \times 1/2 \times 1/2 = 1/16$.
 - For $ppyyrr$: $1/4 \times 1/2 \times 1/2 = 1/16$.
 - Therefore, the chance that a given offspring will have at least two recessive traits is $1/16 + 1/16 + 2/16 + 1/16 + 1/16 = 6/16$.
- Although we cannot predict with certainty the genotype or phenotype of any particular seed from the F_2 generation of a dihybrid cross, we can predict the probability that it will have a specific genotype or phenotype.

Probability laws govern Mendelian inheritance

- Mendel's experiments succeeded because he counted so many offspring, was able to discern the statistical nature of inheritance, and had a keen sense of the rules of chance.
- Mendel's laws of independent assortment and segregation explain heritable variation in terms of alternative forms of genes that are passed along according to simple rules of probability.
- These laws apply not only to garden peas, but to all diploid organisms that reproduce by sexual reproduction.
- Mendel's studies of pea inheritance are a model in genetics and also as a case study of the power of scientific reasoning using the hypothetico-deductive approach.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

- In the 20th century, geneticists extended Mendelian principles both to diverse organisms and to patterns of inheritance more complex than Mendel described.
- In fact, Mendel had the good fortune to choose **a system that was relatively simple genetically**.
 - Each character that Mendel studied is controlled by a single gene. (There is one exception: Mendel's pod shape character is determined by two genes.)
 - Each gene has only two alleles, one of which is completely dominant to the other.
 - The heterozygous F_1 offspring of Mendel's crosses always looked like one of the parental varieties because one allele was dominant to the other.
- **The relationship between genotype and phenotype is rarely so simple.**
- The inheritance of characters determined by a single gene deviates from simple Mendelian patterns when **alleles are not completely dominant or recessive**, when **a gene has more than two alleles**, or when **a gene produces multiple phenotypes**. We will consider each of these situations.

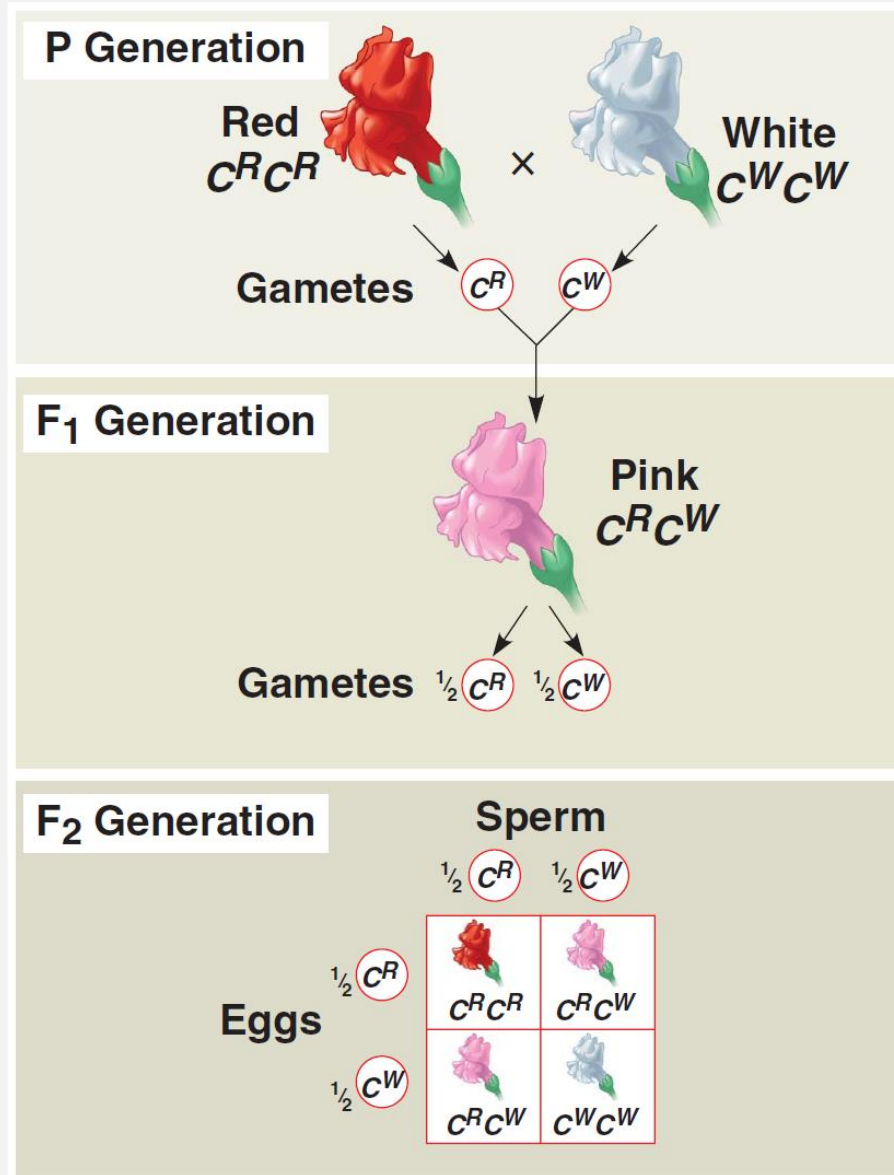
Inheritance patterns are often more complex than predicted by simple Mendelian genetics

There are degrees of dominance.

- Alleles show different degrees of dominance and recessiveness in relation to each other.
- One extreme is the **complete dominance** characteristic of Mendel's crosses.
- Some alleles show **incomplete dominance**, in which heterozygotes show a distinct intermediate phenotype not seen in homozygotes.
 - This is not blending inheritance because the traits are separable (particulate), as shown in further crosses.
 - Offspring of a cross between heterozygotes show three phenotypes: each parental phenotype and the heterozygous phenotype.
 - The phenotypic and genotypic ratios are identical: 1:2:1.
- A clear example of incomplete dominance is the **flower color of snapdragons**.
 - A cross between a white-flowered plant and a red-flowered plant produces all pink F_1 offspring.
 - Self-pollination of the F_1 offspring produces 25% white, 25% red, and 50% pink F_2 offspring.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

There are degrees of dominance.



Inheritance patterns are often more complex than predicted by simple Mendelian genetics

There are degrees of dominance.

- At the other extreme from complete dominance is **codominance**, in which two alleles affect the phenotype in separate, distinguishable ways.
 - For example, the M, N, and MN blood groups of humans are due to the presence of two specific molecules on the surface of red blood cells.
 - People of group M (genotype *MM*) have one type of molecule on their red blood cells, people of group N (genotype *NN*) have the other type, and people of group *MN* (genotype *MN*) have both molecules present.
 - The *MN* phenotype is not intermediate between M and N phenotypes but rather exhibits both the M and the N phenotype.
- The relative effects of two alleles range from complete dominance of one allele, through incomplete dominance of either allele, to codominance of both alleles.
- It is important to recognize that an allele is called *dominant* because it is seen in the phenotype, not because it somehow subdues a recessive allele.
 - Alleles are simply variations in a gene's nucleotide sequence.
 - When a dominant allele coexists with a recessive allele in a heterozygote, they do not interact at all.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

There are degrees of dominance.

- To illustrate the relationship between dominance and phenotype, let's consider Mendel's character of round versus wrinkled pea seed shape.
 - The dominant allele (round) codes for an enzyme that helps convert an unbranched form of starch to a branched form in the seed.
 - The recessive allele (wrinkled) codes for a defective form of this enzyme that leads to an accumulation of unbranched starch.
 - Excess water is then drawn into the seed due to osmosis and the seeds wrinkle when the excess water dries.
 - Both homozygous dominant and heterozygous pea plants produce enough enzymes to synthesize adequate amounts of branched starch.
 - As a result, they do not fill with excess water and they form smooth seeds as they dry.
- For any character, dominance/recessiveness relationships depend on the level at which we examine the phenotype.
 - For example, humans with **Tay-Sachs disease** lack a functioning enzyme to metabolize certain lipids. These lipids accumulate in the brain, harming brain cells and ultimately leading to death.
- Children with two Tay-Sachs alleles (homozygotes) have the disease.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

There are degrees of dominance.

- Children with two Tay-Sachs alleles (homozygotes) have the disease.
- Both heterozygotes, with one allele coding for a functional enzyme, and homozygotes, with two such alleles, are healthy and normal.
 - Thus at the *organismal* level, the allele for the functional enzyme is dominant to the Tay-Sachs allele.
- The activity level of the lipid-metabolizing enzyme is reduced in heterozygotes.
 - Thus, at the *biochemical* level, the alleles show incomplete dominance.
- Heterozygous individuals produce equal numbers of normal and dysfunctional enzyme molecules.
 - At the *molecular* level, the Tay-Sachs and functional alleles are codominant.
- **A dominant allele is not necessarily more common in a population than the recessive allele.**
 - For example, one baby in 400 is born with polydactyly, a condition in which individuals are born with extra fingers or toes.
 - Polydactyly is due to a dominant allele. Clearly, however, the recessive allele is far more prevalent than the dominant allele.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Many genes exist in populations in more than two allelic forms.

- The ABO blood groups in humans are determined by three alleles: I^A , I^B , and i .
 - Both the I^A and I^B alleles are dominant to the i allele.
 - The I^A and I^B alleles are codominant to each other.
- Because each individual carries two alleles, there are six possible genotypes and four possible blood types.
 - Individuals who are $I^A I^A$ or $I^A i$ are type A and have type A carbohydrates on the surface of their red blood cells.
 - Individuals who are $I^B I^B$ or $I^B i$ are type B and have type B carbohydrates on the surface of their red blood cells.
 - Individuals who are $I^A I^B$ are type AB and have both type A and type B carbohydrates on the surface of their red blood cells.
 - Individuals who are ii are type O and have neither carbohydrate on the surface of their red blood cells.
- Matching compatible blood groups is critical for blood transfusions because a person produces antibodies against foreign blood factors.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Many genes exist in populations in more than two allelic forms.

(a) The three alleles for the ABO blood groups and their carbohydrates

Allele	I^A	I^B	i
Carbohydrate	A 	B 	none

(b) Blood group genotypes and phenotypes

Genotype	$I^A I^A$ or $I^A i$	$I^B I^B$ or $I^B i$	$I^A I^B$	ii
Red blood cell with surface carbohydrates				
Phenotype (blood group)	A	B	AB	O

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Most genes are pleiotropic.

- The genes that we have covered so far affect only one phenotypic character.
- Most genes are pleiotropic, affecting more than one phenotypic character.
 - For example, the wide-ranging symptoms of sickle-cell disease are due to a single gene.
- Considering the intricate molecular and cellular interactions responsible for an organism's development, it is not surprising that a gene can affect a number of characteristics.

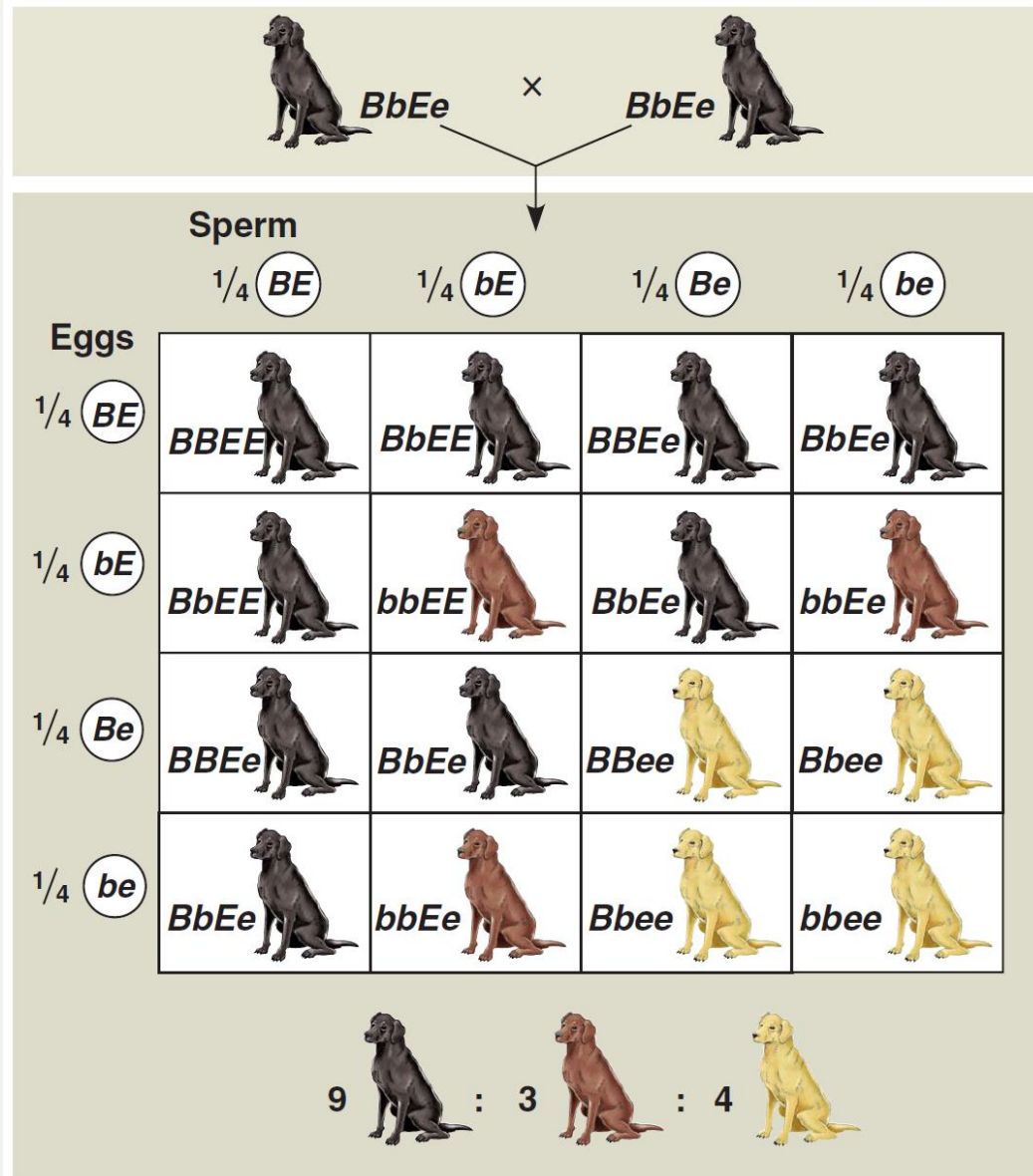
Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Mendelian genetics can be extended for two or more genes.

- Epistasis and polygenic inheritance are situations in which two or more genes are involved in determining phenotype.
- In **epistasis**, the phenotypic expression of a gene at one locus alters the phenotypic expression of a gene at a second locus.
- In **Labrador retrievers**, black coat color is dominant to brown.
 - We'll use B and b as the two alleles for this character. The black allele (B) is dominant to the brown allele (b).
 - Chocolate Labs have brown fur and the genotype b/b .
- A second gene determines whether pigment will be deposited in hair.
 - Presence (E) is dominant to absence (e) of pigment.
 - If the Lab is homozygous recessive at this locus (e/e), its coat is yellow regardless of its genotype at the black/brown locus.
- In this case, the gene for pigment deposition (E/e) is said to be epistatic to the gene that codes for black or brown pigment (B/b).

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Mendelian genetics can be extended for two or more genes.



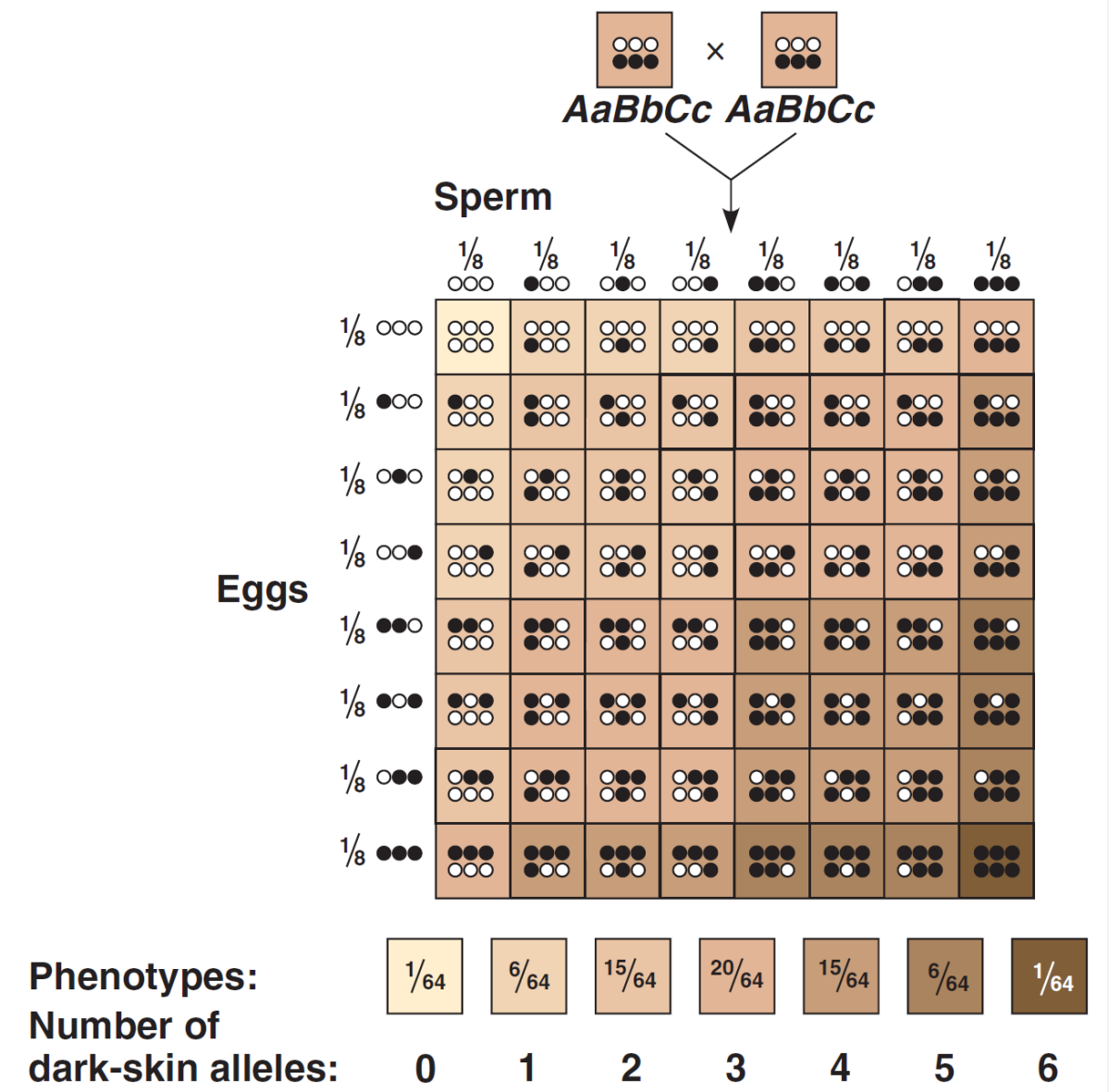
Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Mendelian genetics can be extended for two or more genes.

- A cross between two black Labs that are heterozygous for both genes ($BbEe$) follows the law of independent assortment.
 - Unlike the 9:3:3:1 offspring ratio of a normal Mendelian experiment, however, the offspring ratio is 9 black, 3 chocolate, and 4 yellow.
- Some characters cannot be classified as either-or, as Mendel's genes were.
- Quantitative characters vary in a population along a continuum.
- Quantitative characters are usually due to polygenic inheritance, the additive effects of two or more genes on a single phenotypic character.
- Let's assume that skin color in humans is controlled by three independent genes.
 - Imagine that each gene has two alleles, one light and one dark, which demonstrate incomplete dominance.
 - An $AABBCC$ individual is very dark; an $aabbcc$ individual is very light.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Mendelian genetics can be extended for two or more genes.



Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Mendelian genetics can be extended for two or more genes.

- A cross between two $AaBbCc$ individuals (with intermediate skin shade) produces offspring with a wide range of shades.
 - Individuals with intermediate skin shades are most common, but some very light and very dark individuals may be produced as well.
 - The range of phenotypes forms a normal distribution if the number of offspring is great enough.
 - The majority of offspring would have intermediate phenotypes (skin color in the middle range).
 - Environmental factors, such as sun exposure, also affect skin color and contribute to a smooth normal distribution.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

Genes and environment determine phenotype.

- Phenotype depends on both environment and genes.
 - **A single tree** may have leaves that vary in size, shape, and greenness, depending on exposure to wind and sun.
 - **For humans**, nutrition influences height, exercise alters build, sun-tanning darkens skin, and experience improves performance on intelligence tests.
 - Even identical twins, who are genetically identical, accumulate phenotypic differences as a result of their unique experiences.
- The relative importance of genes and the environment (nature vs. nurture) in influencing human characteristics is a very old and hotly contested debate.
- The product of a genotype is generally not a rigidly defined phenotype, but a range of phenotypic possibilities, the **norm of reaction**, determined by the environment.
 - In some cases, the norm of reaction has no breadth, and a given genotype specifies a particular phenotype (for example, blood type).
 - In contrast, a person's red and white blood cell count varies with factors such as altitude, customary amount of exercise, and presence of infection.
- Norms of reaction are broadest for polygenic characters.
 - For these **multifactorial** characters, genes and environment influence phenotype.

Inheritance patterns are often more complex than predicted by simple Mendelian genetics

A comprehensive theory of Mendelian genetics views organisms as a whole.

- A reductionist emphasis on single genes and single phenotypic characters presents an inadequate perspective on heredity and variation.
- The term **phenotype** can refer not only to specific characters such as flower color or blood group, but also to an organism in its entirety, including *all* aspects of its physical appearance, internal anatomy, physiology, and behavior.
- **Genotype** can refer not only to a single genetic locus but also to an organism's entire genetic makeup.
- An organism's phenotype reflects its overall genotype and its unique environmental history.



(a) Hydrangeas grown in basic soil



(b) Hydrangeas of the same genetic variety grown in acidic soil with free aluminum

Many human traits follow Mendelian patterns of inheritance

- While peas are convenient subjects for genetic research, humans are not.
 - The human generation time is too long, our fecundity is too low, and breeding experiments are unacceptable.
 - Yet, humans are subject to the same rules governing inheritance as other organisms.
- New molecular biology techniques have led to many breakthrough discoveries in the study of human genetics.

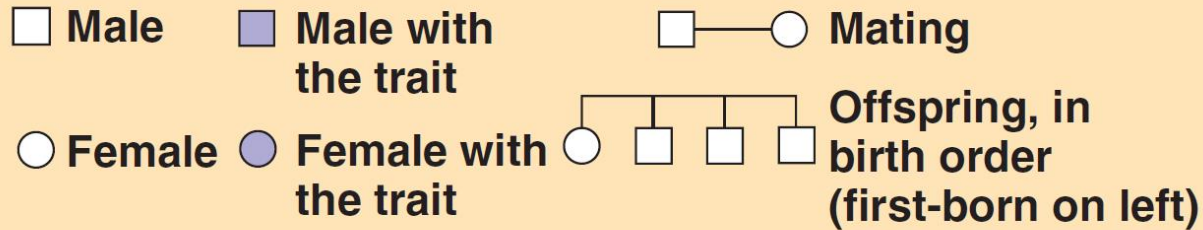
Many human traits follow Mendelian patterns of inheritance

Inheritance of specific traits can be mapped on a family tree.

- In **pedigree analysis**, rather than manipulate mating patterns of people, geneticists analyze the results of matings that have already occurred.
- Information about the presence or absence of a particular phenotypic trait is collected from as many individuals in a family as possible, across generations.
- The distribution of these characters is then mapped on the family tree.
 - For example, the occurrence of a widow's peak (W) is dominant to a straight hairline (w).
- Phenotypes of family members and knowledge of dominance/recessiveness relationships between alleles allow researchers to predict the genotypes of members of this family.
 - If a sibling in the second generation lacks a widow's peak and one of the grandparents (first generation) also lacks one, then we know the other grandparent must be heterozygous.
 - If an individual in the third generation lacks a widow's peak but both her parents have widow's peaks, then her parents must be heterozygous for that gene.
 - From these facts, we can determine the genotype of other individuals.

Many human traits follow Mendelian patterns of inheritance

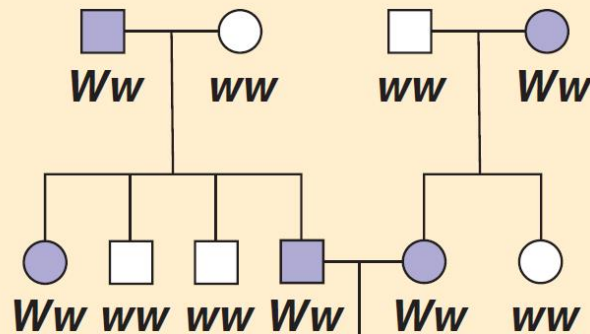
Inheritance of specific traits can be mapped on a family tree.



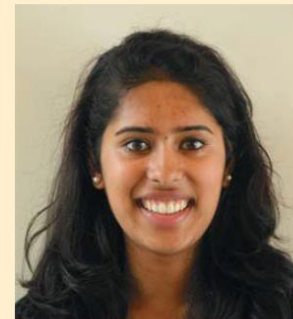
1st generation
(grandparents)

2nd generation
(parents, aunts,
and uncles)

3rd generation
(two sisters)



Widow's peak

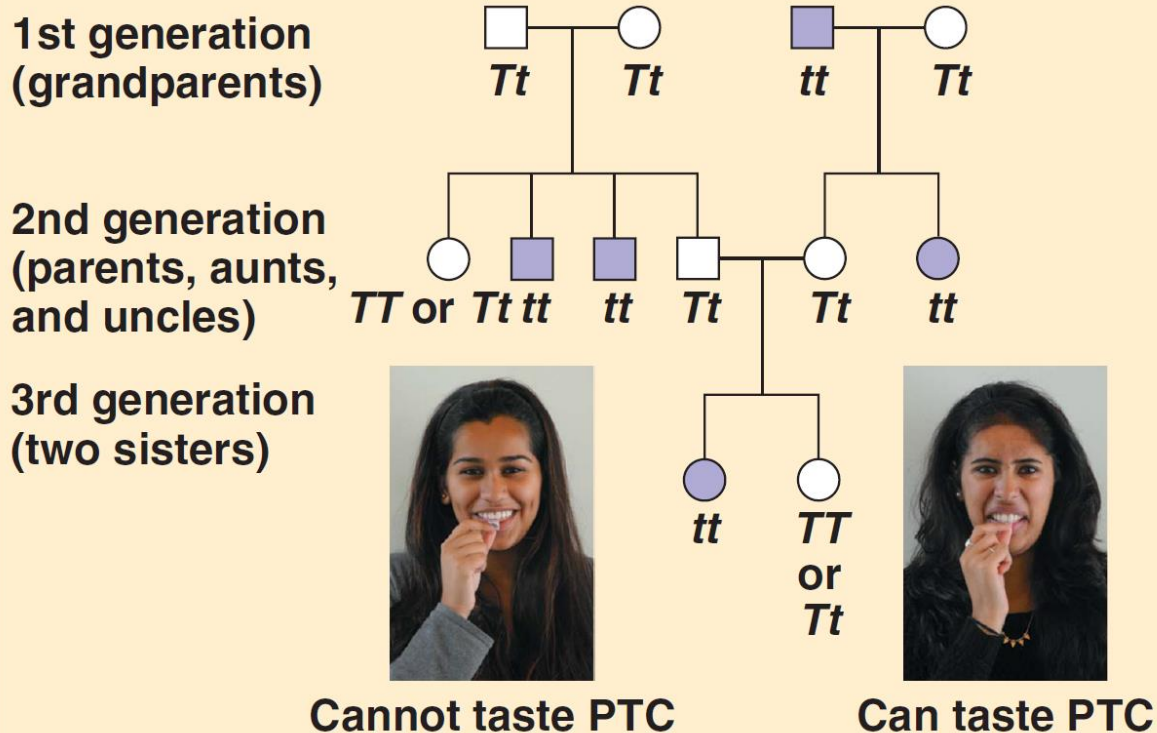
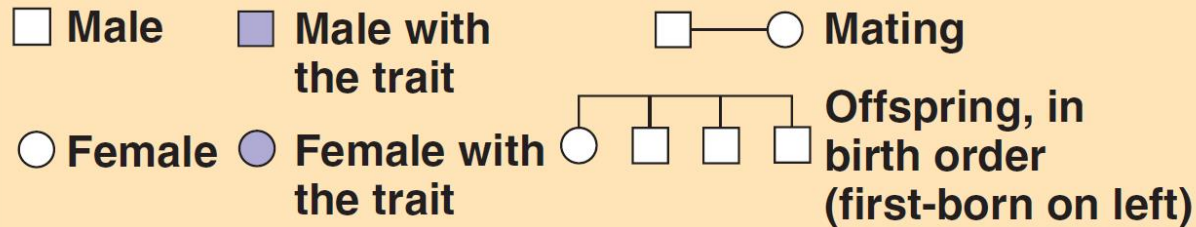


No widow's peak

(a) Is a widow's peak a dominant or recessive trait?

Many human traits follow Mendelian patterns of inheritance

Inheritance of specific traits can be mapped on a family tree.



(b) Is the inability to taste a chemical called PTC a dominant or recessive trait?

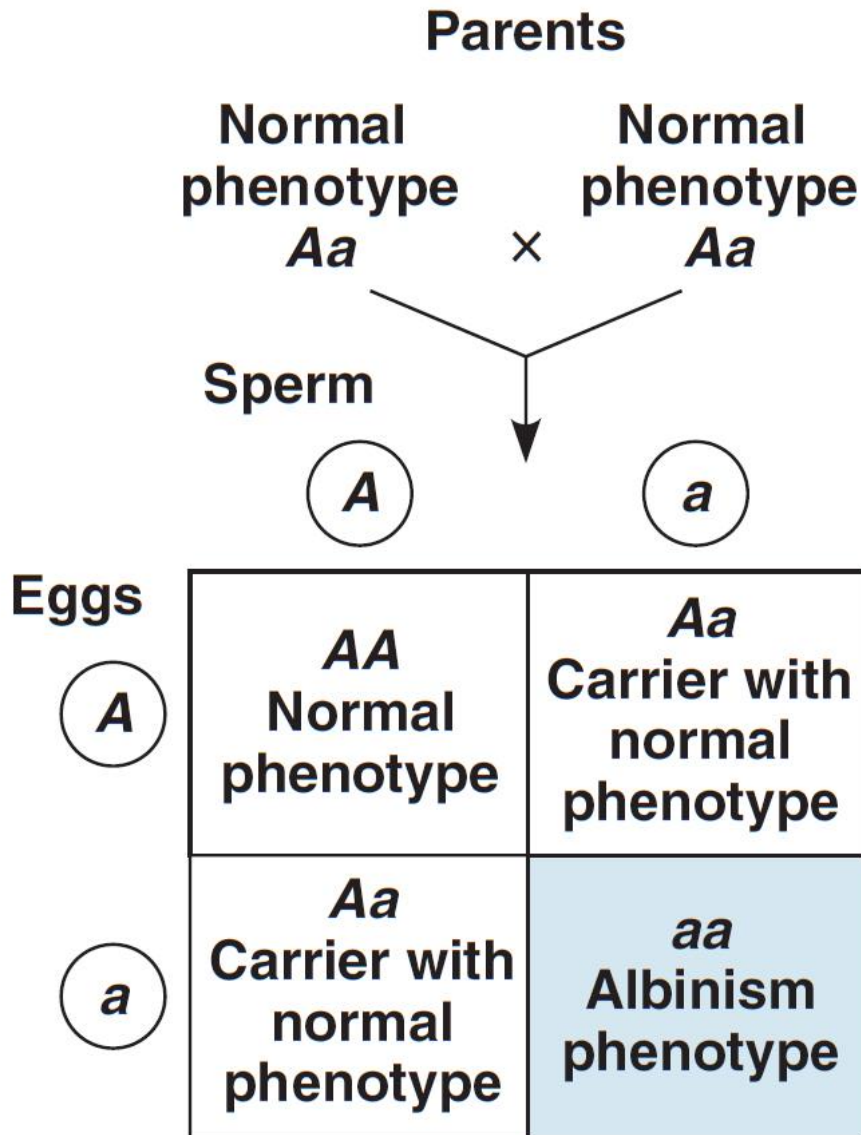
Many human traits follow Mendelian patterns of inheritance

Some human disorders are recessively inherited.

- Thousands of genetic disorders, including disabling or deadly hereditary diseases, are inherited as simple recessive traits.
 - These conditions range from relatively mild (albinism) to life-threatening (cystic fibrosis).
- An allele that causes a recessive condition such as albinism codes for a malfunctioning protein or for no protein at all.
- Heterozygotes have a normal phenotype because *one normal allele produces enough of the required protein*.
- Albinism shows up only in homozygous individuals who inherit a recessive allele from each parent.
- Individuals who do not have the disorder are either homozygous dominant or heterozygotes.
- Although heterozygotes may lack obvious phenotypic effects, they are **carriers** who may transmit a recessive allele to their offspring.
- Most people with recessive disorders are born to carriers with normal phenotypes.
 - In a mating between two carriers of albinism, each child has a 1/4 chance of inheriting the disorder, a 1/2 chance of being a carrier, and a 1/4 chance of being homozygous dominant.

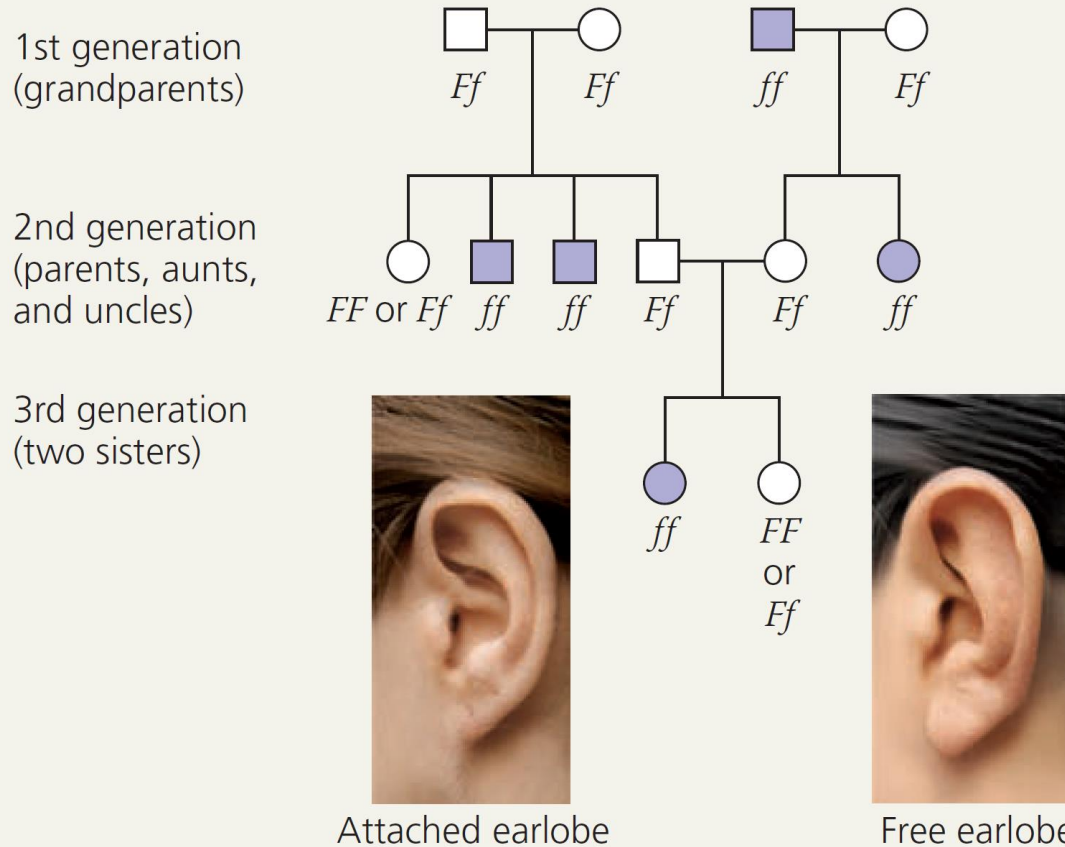
Many human traits follow Mendelian patterns of inheritance

Some human disorders are recessively inherited.



Many human traits follow Mendelian patterns of inheritance

Some human disorders are recessively inherited.



(b) Is an attached earlobe a dominant or recessive trait?

Tips for pedigree analysis: Notice that the first-born daughter in the third generation has attached earlobes, although both of her parents lack that trait (they have free earlobes). Such a pattern is easily explained if the attached-lobe phenotype is due to a recessive allele. If it were due to a *dominant* allele, then at least one parent would also have had the trait.

Many human traits follow Mendelian patterns of inheritance

Some human disorders are recessively inherited.

- Genetic disorders are not evenly distributed among all groups of humans.
 - This is due to the different genetic histories of the world's people during times when populations were more geographically and genetically isolated.
- Normally, it is relatively unlikely that two carriers of the same rare, harmful allele will meet and mate.
- Consanguineous matings between close relatives increase the risk.
 - Individuals who share a recent common ancestor are more likely to carry the same recessive alleles and to produce offspring homozygous for recessive traits—including harmful ones.
 - Geneticists disagree about the extent to which human consanguinity increases the risk of inherited diseases.
- Most societies and cultures have laws or taboos forbidding marriages between close relatives.

Many human traits follow Mendelian patterns of inheritance

Some human disorders are recessively inherited.

- **Cystic fibrosis** strikes one of every 2,500 whites of European descent.
 - One in 25 people (4%) of European descent is a carrier for this condition.
 - The normal allele at this gene codes for a membrane protein that transports chloride between cells and extracellular fluid.
 - If these channels are defective or absent, abnormally high extracellular levels of chloride accumulate.
 - Then the mucous coats of certain cells become thicker and stickier than normal.
 - This mucous buildup in the pancreas, lungs, digestive tract, and elsewhere causes *poor absorption of nutrients, chronic bronchitis, and bacterial infections*.
- Without treatment, affected children die before age 5, but with treatment, they can live much longer.

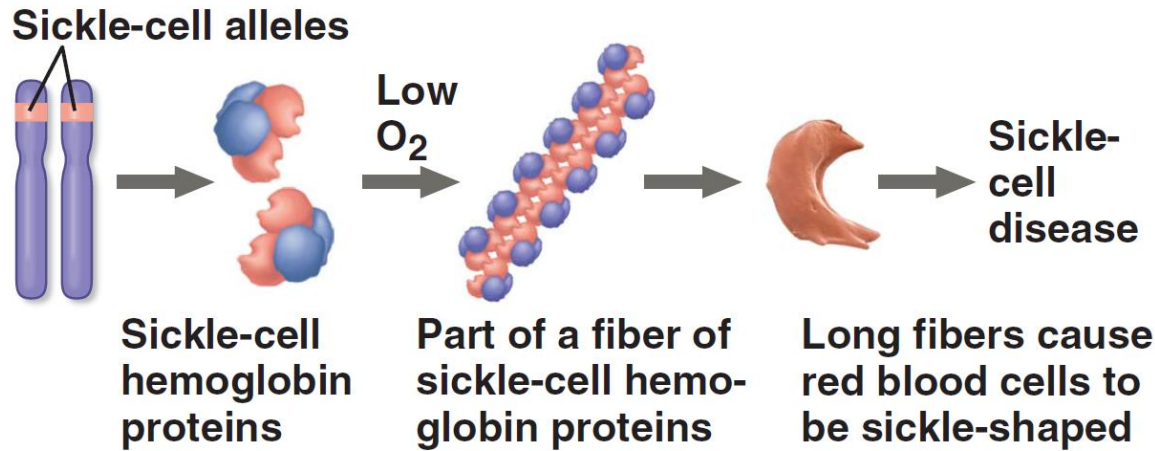
Many human traits follow Mendelian patterns of inheritance

Some human disorders are recessively inherited.

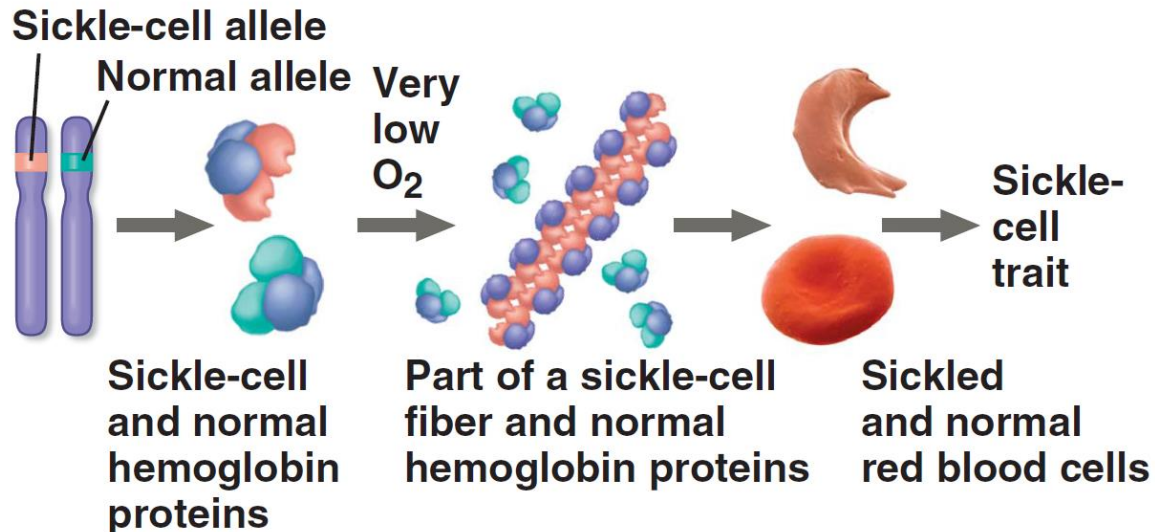
- The most common inherited disease among people of African descent is **sickle-cell disease**, which affects one of 400 African-Americans.
 - Sickle-cell disease is caused by the substitution of a single amino acid in hemoglobin; in homozygous individuals, all hemoglobin is of the sickle-cell (abnormal) variety.
 - When oxygen levels in the blood of an affected individual are low, sickle-cell hemoglobin aggregates into long rods that deform red blood cells into a sickle shape.
 - This sickling creates a cascade of symptoms, demonstrating **the pleiotropic effects of this allele**, as sickled cells clump and clog capillaries throughout the body.
- Doctors can use regular blood transfusions to prevent brain damage and new drugs to prevent or treat other problems.
- **At the organismal level, the nonsickle allele is incompletely dominant to the sickle-cell allele.**
 - Carriers are said to have *sickle-cell trait*.
 - These individuals are usually healthy, although some may suffer symptoms of sickle-cell disease under blood-oxygen stress.
- **At the molecular level, the two alleles are codominant because both normal and abnormal (sickle-cell) hemoglobin molecules are synthesized.**

Many human traits follow Mendelian patterns of inheritance

Some human disorders are recessively inherited.



(a) Homozygote with sickle-cell disease: Weakness, anemia, pain and fever, organ damage



(b) Heterozygote with sickle-cell trait: Some symptoms when blood oxygen is very low; reduction of malaria symptoms

Many human traits follow Mendelian patterns of inheritance

Some human disorders are recessively inherited.

- About one in ten African-Americans has sickle-cell trait.
 - The high frequency of heterozygotes is unusual for an allele with severe detrimental effects in homozygotes.
- Individuals with one sickle-cell allele have increased resistance to the malaria parasite, which spends part of its life cycle in red blood cells.
 - In tropical Africa, where malaria is common, the sickle-cell allele is both a boon and a bane.
 - Homozygous normal individuals die of malaria and homozygous recessive individuals die of sickle-cell disease, while carriers are relatively free of both.
- The relatively high frequency of sickle-cell trait in African-Americans is a vestige of their African roots.

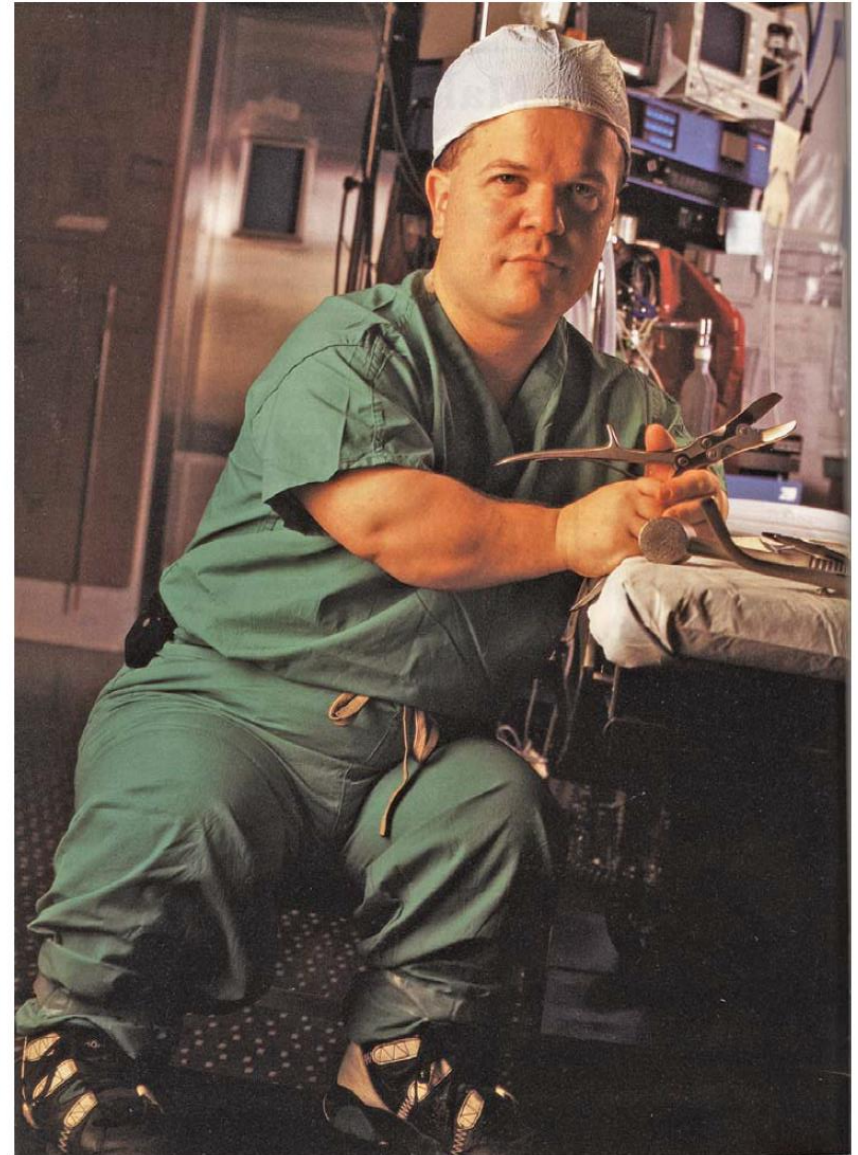
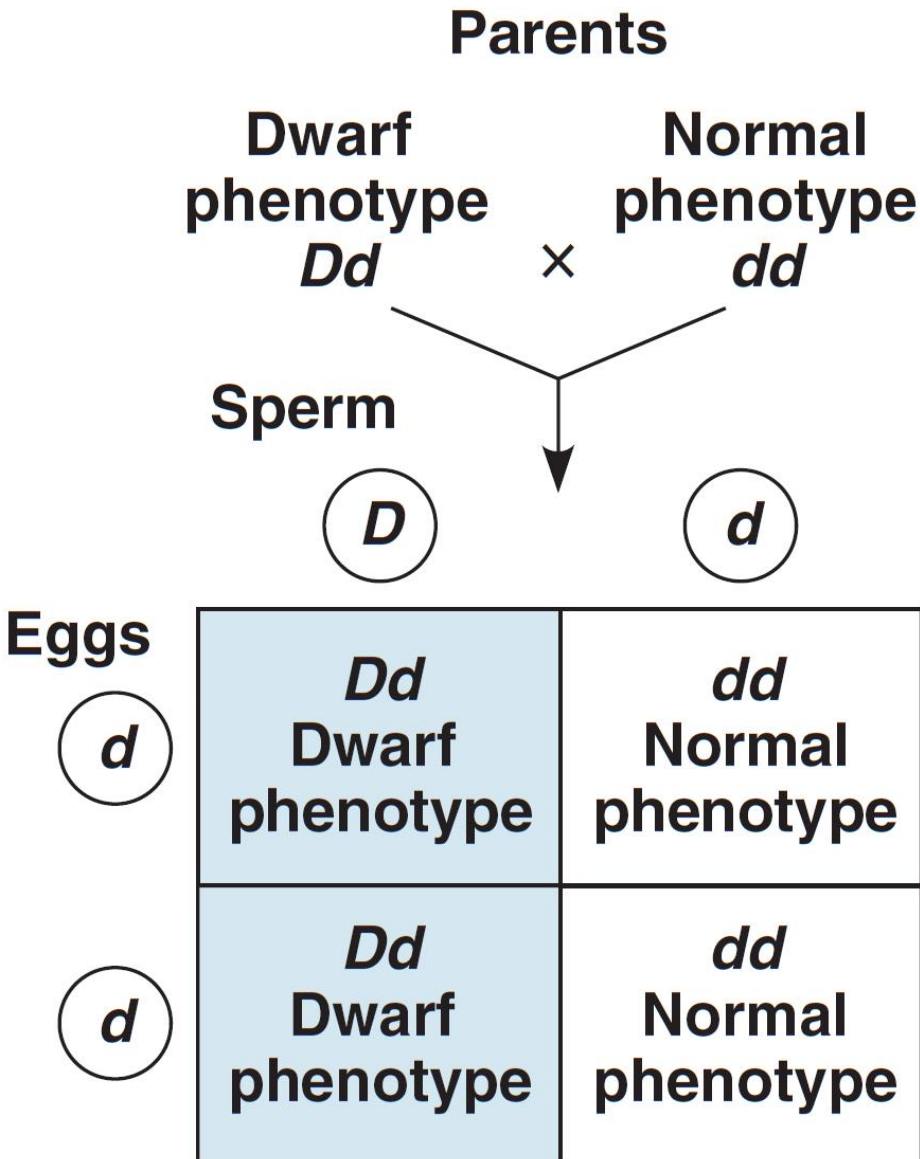
Many human traits follow Mendelian patterns of inheritance

Some human disorders are dominantly inherited.

- Although most harmful alleles are recessive, a number of human disorders are due to dominant alleles.
- ***Achondroplasia***, a form of dwarfism, has a prevalence of one case in 25,000 people.
 - Heterozygous individuals have the dwarf phenotype.
- The 99.99% of the population who are not achondroplastic dwarfs are homozygous recessive for this trait.
 - Achondroplasia is another example of a trait for which the recessive allele is far more prevalent than the dominant allele.
- **Lethal dominant alleles are much less common than lethal recessive alleles.**
 - If a lethal dominant allele kills an offspring before he or she can mature and reproduce, the allele will not be passed on to future generations.
 - In contrast, a lethal recessive allele can be passed on by heterozygous carriers who have normal phenotypes.

Many human traits follow Mendelian patterns of inheritance

Some human disorders are dominantly inherited.



Human Traits

Many human traits follow Mendelian patterns of inheritance

Some human disorders are dominantly inherited.

- *A lethal dominant allele can escape elimination if it causes death at a relatively advanced age, after the individual has already passed on the lethal allele to his or her children.*
- One example is **Huntington's disease**, a degenerative disease of the nervous system.
 - The dominant lethal allele has no obvious phenotypic effect until the individual is about 35 to 45 years old.
 - Then the deterioration of the nervous system is irreversible and inevitably fatal.
- Any child born to a parent who has the allele for Huntington's disease has a 50% chance of inheriting the disease and the disorder.
- In the United States, this devastating disease afflicts one in 10,000 people.
- **Molecular geneticists used pedigree analysis of affected families to track the Huntington's allele to a locus near the tip of chromosome 4.**
 - The gene was sequenced in 1993, leading to the development of a test that can detect the presence of the Huntington's allele in an individual's genome.

Many human traits follow Mendelian patterns of inheritance

Some disorders are multifactorial.

- While some diseases are inherited in a simple Mendelian fashion due to alleles at a single locus, many other disorders have a multifactorial basis.
 - Such disorders have a genetic component plus a significant environmental influence.
 - Multifactorial disorders include **heart disease, diabetes, cancer, alcoholism, and certain mental illnesses, such as schizophrenia and manic-depressive disorder.**
 - The genetic component of such disorders is typically polygenic.
- At present, little is understood about the genetic contribution to most multifactorial diseases.
 - The best public health strategy is education about relevant environmental factors and promotion of healthy behavior.

Many human traits follow Mendelian patterns of inheritance

Genetic counseling is based on Mendelian genetics and the laws of probability.

- A preventive approach to simple Mendelian disorders is sometimes possible.
- The risk that a particular genetic disorder will occur can sometimes be assessed before a child is conceived or early in pregnancy.
- Many hospitals have genetic counselors to provide information to prospective parents who are concerned about a family history of a specific disease.
- **Consider a hypothetical couple, John and Carol, who are planning to have their first child.**
- Both John and Carol had brothers who died of the same recessive disease.
 - John, Carol, and their parents do not have the disease. Their parents must have been carriers ($Aa \times Aa$).
 - John and Carol each have a $2/3$ chance of being carriers and a $1/3$ chance of being homozygous dominant.
 - The probability that their first child will have the disease is $2/3$ (chance that John is a carrier) $\times$ $2/3$ (chance that Carol is a carrier) $\times$ $1/4$ (chance that the offspring of two carriers is homozygous recessive) = $1/9$.
- If their first child is born with the disease, we know that John and Carol's genotype must be Aa and they are both carriers.
 - In that case, the chance that their next child will also have the disease is $1/4$.
 - The probability is higher because the diagnosis of the disease in the first child established that both parents are carriers, not because the genotype of the first child affects in any way that of future children.

Many human traits follow Mendelian patterns of inheritance

Genetic counseling is based on Mendelian genetics and the laws of probability.

- Mendel's laws are simply the rules of probability applied to heredity.
 - Because chance has no memory, the genotype of each child is unaffected by the genotypes of older siblings.
 - The chance that John and Carol's first three children will have the disorder is $1/4 \times 1/4 \times 1/4 = 1/64$.
 - Should that outcome happen, the likelihood that a fourth child will also have the disorder is still $1/4$.
- Because most children with recessive disorders are born to parents with a normal phenotype, the key to assessing risk is identifying whether prospective parents are heterozygous carriers of the recessive trait.
- Recently developed tests for many disorders can distinguish normal phenotypes in heterozygotes from homozygous dominants.
 - These results allow individuals with a family history of a genetic disorder to make informed decisions about having children.
 - Issues of confidentiality, discrimination, and counseling may arise.
 - The Genetic Information Nondiscrimination Act, signed into US law in 2008, prohibits discrimination in employment or insurance coverage based on genetic test results.
- Tests are available to determine *in utero* whether a child has a particular disorder.

Many human traits follow Mendelian patterns of inheritance

Genetic counseling is based on Mendelian genetics and the laws of probability.

- One technique, **amniocentesis**, can be used from the 14th to 16th week of pregnancy to assess whether the fetus has a specific disease.
 - Fetal cells extracted from amniotic fluid are cultured and karyotyped to identify some disorders.
 - Other disorders can be identified from chemicals in the amniotic fluids.
- A second technique, **chorionic villus sampling (CVS)**, allows faster karyotyping and can be performed as early as the 8th to 10th week of pregnancy.
 - A sample of fetal tissue is extracted from the chorionic villi of the placenta.
- Recent techniques allow the isolation of fetal cells or DNA from the mother's blood.
- Other techniques allow fetal health to be assessed visually *in utero*.
 - *Ultrasound* uses sound waves to produce an image of the fetus.
 - In *fetoscopy*, a needle-thin tube containing fiber optics and a viewing scope is inserted into the uterus.
- Ultrasound and isolation of fetal cells or DNA from maternal blood pose no known risk to mother or fetus, while the other procedures have a small percentage of complications.
- If fetal tests reveal a serious disorder, the parents face the difficult choice of terminating the pregnancy or preparing to care for a child with a genetic disorder.

Many human traits follow Mendelian patterns of inheritance

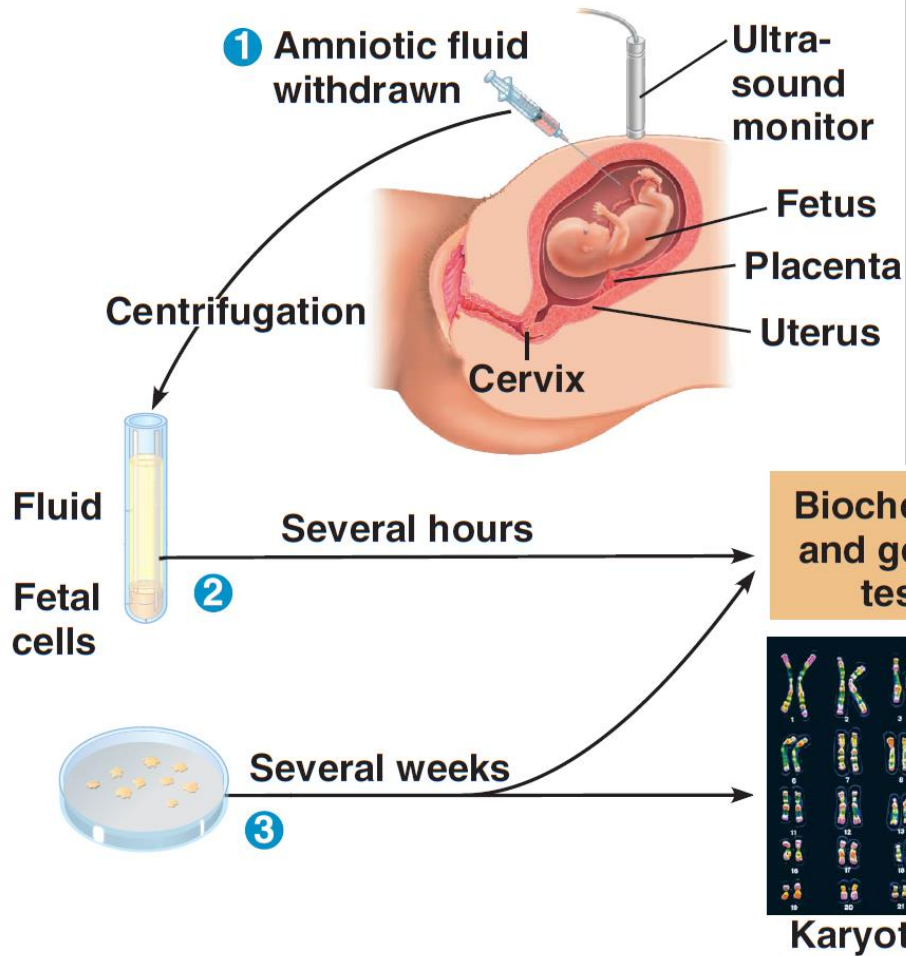
Genetic counseling is based on Mendelian genetics and the laws of probability.

- Some genetic traits can be detected at birth by simple tests that are now routinely performed in hospitals.
- One test can detect the presence of a recessively inherited disorder, phenylketonuria (PKU).
 - This disorder occurs in one in 10,000 to 15,000 births.
 - Individuals with this disorder accumulate the amino acid phenylalanine and its derivative phenylpyruvate in the blood to toxic levels, which leads to *mental retardation*.
 - If the disorder is detected, a **special diet low in phenylalanine** usually promotes normal development.
- Unfortunately, few other genetic diseases are treatable at present.

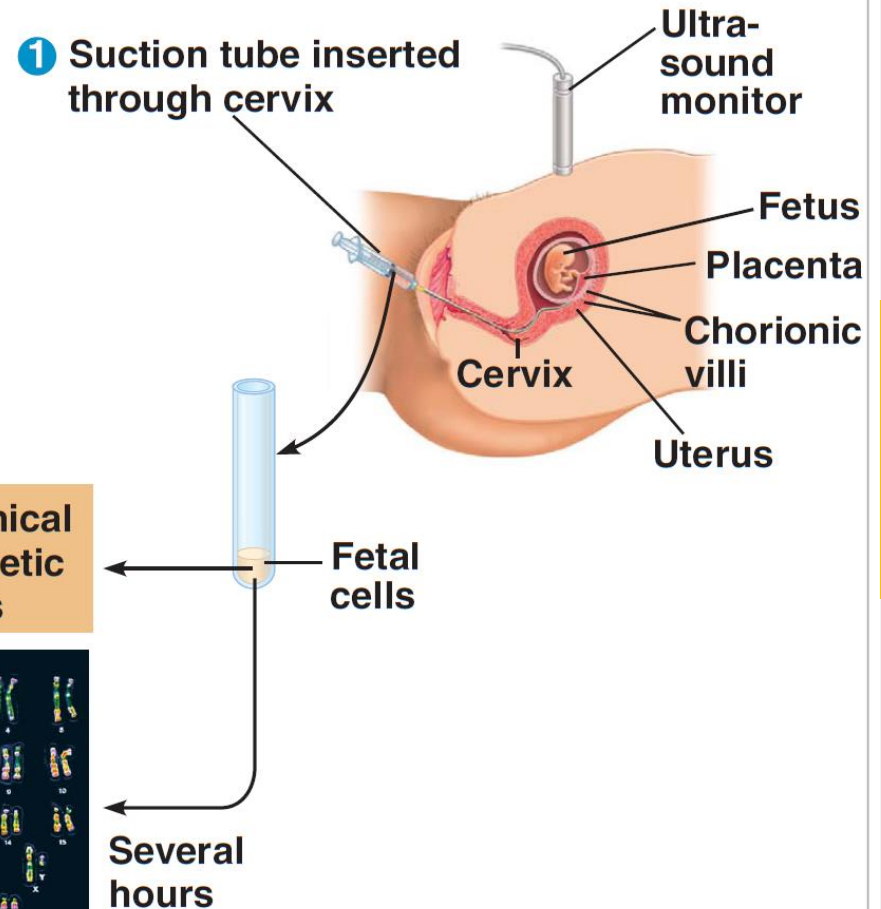
Many human traits follow Mendelian patterns of inheritance

Genetic counseling is based on Mendelian genetics and the laws of probability.

(a) Amniocentesis



(b) Chorionic villus sampling (CVS)



SUMMARY OF KEY CONCEPTS

CONCEPT 14.1

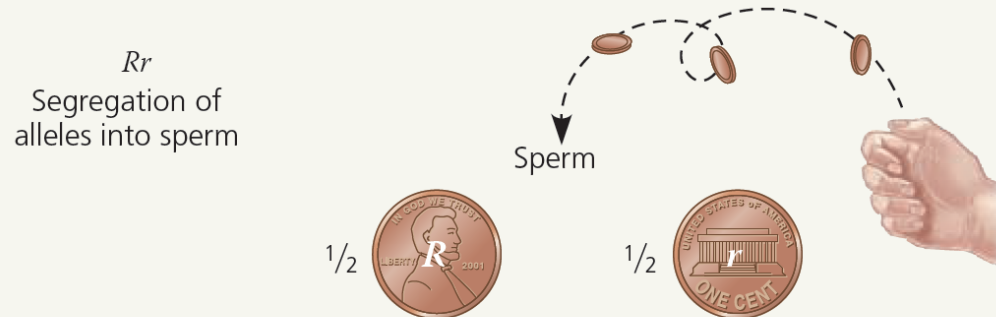
Mendel used the scientific approach to identify two laws of inheritance (pp. 268–274)

- Gregor Mendel formulated a theory of inheritance based on experiments with garden peas, proposing that parents pass on to their offspring discrete genes that retain their identity through generations. This theory includes two “laws.”
- The **law of segregation** states that genes have alternative forms, or **alleles**. In a diploid organism, the two alleles of a gene segregate (separate) during meiosis and gamete formation; each sperm or egg carries only one allele of each pair. This law explains the 3:1 ratio of F_2 phenotypes observed when **monohybrids** self-pollinate. Each organism inherits one allele for each gene from each parent. In **heterozygotes**, the two alleles are different; expression of the **dominant allele** masks the phenotypic effect of the **recessive allele**. **Homozygotes** have identical alleles of a given gene and are **true-breeding**.
- The **law of independent assortment** states that the pair of alleles for a given gene segregates into gametes independently of the pair of alleles for any other gene. In a cross between **dihybrids** (individuals heterozygous for two genes), the offspring have four phenotypes in a 9:3:3:1 ratio.

? When Mendel did crosses of true-breeding purple- and white-flowered pea plants, the white-flowered trait disappeared from the F_1 generation but reappeared in the F_2 generation. Use genetic terms to explain why that happened.

CONCEPT 14.2

Probability laws govern Mendelian inheritance (pp. 274–276)



- The **multiplication rule** states that the probability of two or more events occurring together is equal to the product of the individual probabilities of the independent single events. The **addition rule** states that the probability of an event that can occur in two or more independent, mutually exclusive ways is the sum of the individual probabilities.
- The rules of probability can be used to solve complex genetics problems. A dihybrid or other multicharacter cross is equivalent to two or more independent monohybrid crosses occurring simultaneously. In calculating the chances of the various offspring genotypes from such crosses, each character is first considered separately and then the individual probabilities are multiplied.

DRAW IT Redraw the Punnett square on the right side of Figure 14.8 as two smaller monohybrid Punnett squares, one for each gene. Below each square, list the fractions of each phenotype produced. Use the rule of multiplication to compute the overall fraction of each possible dihybrid phenotype. What is the phenotypic ratio?

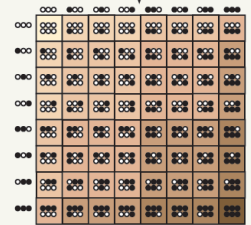
CONCEPT 14.3

Inheritance patterns are often more complex than predicted by simple Mendelian genetics (pp. 276–281)

- Extensions of Mendelian genetics for a single gene:

Relationship among alleles of a single gene	Description	Example
Complete dominance of one allele	Heterozygous phenotype same as that of homozygous dominant	PP  Pp 
Incomplete dominance of either allele	Heterozygous phenotype intermediate between the two homozygous phenotypes	$C^R C^R$  $C^R C^W$  $C^W C^W$ 
Codominance	Both phenotypes expressed in heterozygotes	$I^A I^B$ 
Multiple alleles	In the population, some genes have more than two alleles	ABO blood group alleles I^A, I^B, i
Pleiotropy	One gene affects multiple phenotypic characters	Sickle-cell disease

- Extensions of Mendelian genetics for two or more genes:

Relationship among two or more genes	Description	Example
Epistasis	The phenotypic expression of one gene affects the expression of another gene	$BbEe$  $\times$ $BbEe$  BE bE Be be BE     bE     Be     be     9  : 3  : 4 
Polygenic inheritance	A single phenotypic character is affected by two or more genes	$AaBbCc$  $\times$ $AaBbCc$  

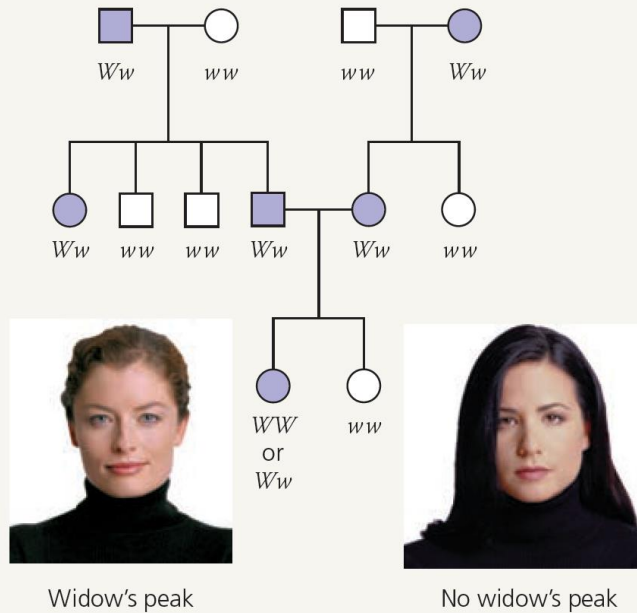
- The expression of a genotype can be affected by environmental influences, resulting in a range of phenotypes. Polygenic characters that are also influenced by the environment are called **multifactorial** characters.
- An organism's overall phenotype, including its physical appearance, internal anatomy, physiology, and behavior, reflects its overall genotype and unique environmental history. Even in more complex inheritance patterns, Mendel's fundamental laws of segregation and independent assortment still apply.

? Which of the following are demonstrated by the inheritance patterns of the ABO blood group alleles: complete dominance, incomplete dominance, codominance, multiple alleles, pleiotropy, epistasis, and/or polygenic inheritance? Explain each of your answers.

CONCEPT 14.4

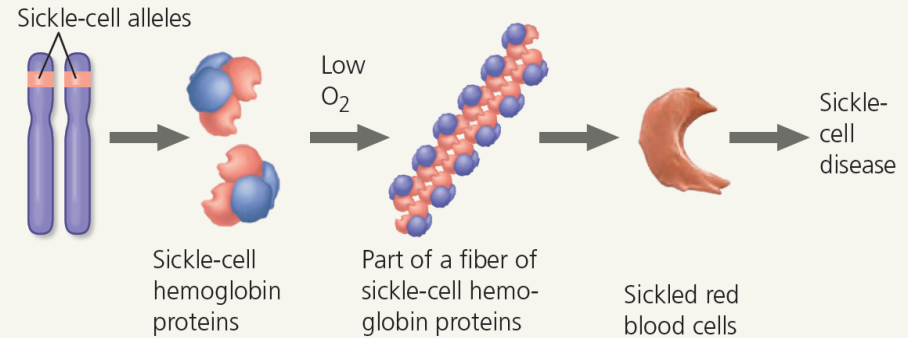
Many human traits follow Mendelian patterns of inheritance (pp. 282–288)

- Analysis of family **pedigrees** can be used to deduce the possible genotypes of individuals and make predictions about future offspring. Such predictions are statistical probabilities rather than certainties.



- Many genetic disorders are inherited as simple recessive traits. Most affected (homozygous recessive) individuals are children of phenotypically normal, heterozygous **carriers**.

- The sickle-cell allele has probably persisted for evolutionary reasons: Heterozygotes have an advantage because one copy of the sickle-cell allele reduces both the frequency and severity of malaria attacks.



- Lethal dominant alleles are eliminated from the population if affected people die before reproducing. Nonlethal dominant alleles and lethal ones that strike relatively late in life can be inherited in a Mendelian way.
- Many human diseases are multifactorial—that is, they have both genetic and environmental components and do not follow simple Mendelian inheritance patterns.
- Using family histories, genetic counselors help couples determine the probability that their children will have genetic disorders. Genetic testing of prospective parents to reveal whether they are carriers of recessive alleles associated with specific disorders has become widely available. **Amniocentesis** and **chorionic villus sampling** can indicate whether a suspected genetic disorder is present in a fetus. Other genetic tests can be performed after birth.

TIPS FOR GENETICS PROBLEMS

1. Write down symbols for the alleles. (These may be given in the problem.) When represented by single letters, the dominant allele is uppercase and the recessive is lowercase.
2. Write down the possible genotypes, as determined by the phenotype.
 - a. If the phenotype is that of the dominant trait (for example, purple flowers), then the genotype is either homozygous dominant or heterozygous (PP or Pp , in this example).
 - b. If the phenotype is that of the recessive trait, the genotype must be homozygous recessive (for example, pp).
 - c. If the problem says “true-breeding,” the genotype is homozygous.
3. Determine what the problem is asking for. If asked to do a cross, write it out in the form [Genotype] $\times$ [Genotype], using the alleles you’ve decided on.
4. To figure out the outcome of a cross, set up a Punnett square.
 - a. Put the gametes of one parent at the top and those of the other on the left. To determine the allele(s) in each gamete for a given genotype, set up a systematic way to list all the possibilities. (Remember, each gamete has one allele of each gene.) Note that there are 2^n possible types of gametes, where n is the number of gene loci that are heterozygous. For example, an individual with genotype $AaBbCc$ would produce $2^3 = 8$ types of gametes. Write the genotypes of the gametes in circles above the columns and to the left of the rows.
 - b. Fill in the Punnett square as if each possible sperm were fertilizing each possible egg, making all of the possible offspring. In a cross of $AaBbCc \times AaBbCc$, for example, the Punnett square would have 8 columns and 8 rows, so there are 64 different offspring; you would know the genotype of each and thus the phenotype. Count genotypes and phenotypes to obtain the genotypic and phenotypic ratios. Because the Punnett square is so large, this method is not the most efficient. See tip 5.
5. You can use the rules of probability if the Punnett square would be too big. (For example, see the question at the end of the summary for Concept 14.2 and question 7 below.) You can consider each gene separately (see Concept 14.2).
6. If, instead, the problem gives you the phenotypic ratios of offspring but not the genotypes of the parents in a given cross, the phenotypes can help you deduce the parents’ unknown genotypes.
 - a. For example, if $\frac{1}{2}$ of the offspring have the recessive phenotype and $\frac{1}{2}$ the dominant, you know that the cross was between a heterozygote and a homozygous recessive.
 - b. If the ratio is 3:1, the cross was between two heterozygotes.
 - c. If two genes are involved and you see a 9:3:3:1 ratio in the offspring, you know that each parent is heterozygous for both genes. Caution: Don’t assume that the reported numbers will exactly equal the predicted ratios. For example, if there are 13 offspring with the dominant trait and 11 with the recessive, assume that the ratio is one dominant to one recessive.
7. For pedigree problems, use the tips in Figure 14.15 and below to determine what kind of trait is involved.
 - a. If parents without the trait have offspring with the trait, the trait must be recessive and the parents both carriers.
 - b. If the trait is seen in every generation, it is most likely dominant (see the next possibility, though).
 - c. If both parents have the trait and the trait is recessive, all offspring will show the trait.
 - d. To determine the likely genotype of a certain individual in a pedigree, first label the genotypes of all the family members you can. Even if some of the genotypes are incomplete, label what you do know. For example, if an individual has the dominant phenotype, the genotype must be AA or Aa ; you can write this as $A-$. Try different possibilities to see which fits the results. Use the rules of probability to calculate the probability of each possible genotype being the correct one.